

LESSON NOTES OF MATHEMATICS FOR P.5 TERM ONE

PRIMARY FIVE MATHEMATICS TOPICAL BREAKDOWN OF LESSON NOTES

Theme	Topic	Sub topic (content)	Duration	Learning outcomes
Sets	Set Concept	- Reviewed sets concepts - Definition - Describing, shading, listing and forming sets - Types of sets and symbols - Intersection and union sets - Equivalent sets and non-equivalent. - Equal and non-equal sets - Empty/Null sets - Subsets (Proper and improper) - Complement of sets - Shading sets on the venn diagram - Sets on venn diagram - Representing sets - Interpreting information from the venn diagram - Probability (chance) - Probability of a coin - Probability of a dice. - Probability of days of the week.		The learner is able to demonstrate the knowledge of the sets to show problems real life situations.
Numeracy	Whole numbers	- Forming numbers from digits (sum and difference) - Values of numbers - Sum, difference and product of value of numbers. - Expanding whole numbers using - Place values - Using values	2weeks	The learner is able to appreciate the need counting everyday life and work with whole numbers up to 999,999

			○ Using exponents / powers of 10 • Expanded numbers • Writing words in words • Writing words into figures • Rounding off whole numbers up to 10,000(ten thousands) • Roman numerals up to 300 (ccc)- Hindu –Arabic numerals			
	Numeracy	• Operations on whole numbers	• addition of whole numbers up to 999,999 - with and without regrouping. • Word problem about addition • Subtraction of whole numbers with and without grouping up to 6 digits • Word problems on subtraction. • Multiplication of whole numbers of 4 digits by 2 digits. • Word problems on multiplication • Division of whole numbers up to 5 digits by 2 digits. - Without and with a remainder.	3 weeks	The learner is able to use the four basic operations to solve problems.	
			• Word problems involving division of whole numbers. • Combined operations. (BODMAS) • Statistics; - Mean - Mode - Median - Range • Decimal and base five (Quandary base) - Expanding in base five - Writing base five numbers in words.			

			- Converting base five to base ten. - Changing base ten to base five. - Addition of numbers in base five. - Subtraction of numbers in base five. • Finite system - Expressing numbers in finite system. - Addition of numbers in finite using a dial and calculus. - Subtraction of numbers infinite system using a dial and calculus. • Application of finite system.			
	Numeracy	• Number and sequence	• Types of numbers - Whole numbers - Counting numbers - Even numbers - Odd numbers - Triangular numbers - Prime numbers - Square numbers (composite numbers) - Cube numbers - Square numbers - Finding square of numbers (neglect square roots) • Multiples of numbers • LCM of numbers by listing multiples. • GCF of numbers by listing factors. • Prime factorization of numbers - By ladder method - By use of factor tree. • Listing prime factors using - Subscript form (set notation) - Power form • Sequences - Increasing progression (addition and multiplication) - Decreasing progression.	2 weeks	The learner is able to relate and apply simple comprehension, skills involving patterns and sequences to real life situations.	

			(subtraction and division)			
	Numera cy	• Fractio ns	• Addition • Subtraction with same and different denominators • Multiplication of fractions - Natural numbers - By proper fractions - By reciprocal • Division of fractions - By proper fractions - By natural numbers and vice versa. - Ordering fractions • Interpreting and solving problems in real life situation about fraction	1½ weeks	The learner is able to solve problems involving fractions and relating them to real life situations.	

TERM ONE: TOPIC ONE

Topic: sets

Sub topic: types of sets

Content: definition of terms

- (a) A set is a well-defined collection of elements or members.
- (b) Union of sets is a collection of elements in 2 or more sets without representing common members.
- (c) Intersection of sets are common elements in 2 or more sets

Types of sets

Equal and equivalent sets e.g. $\{1, 2, 3\}$ B $\{2, 1, 3\}$

Set A = B

Set K = $\{a, b, c\}$ set L = $\{m, n, o\}$

Set K equivalent to L

$$K \Leftrightarrow L$$

Equal and unequal sets

P = $\{5, 4, 6\}$ set N = $\{a, b, c\}$

$$P \neq N$$

Definition of terms

Equal sets (same numbers of elements of same kind)

Equivalent sets (same number of elements of different kinds)

No equivalent sets (different number of elements of different elements)

Examples

(i) $A = \{a, e, I, o, u\}$ $B = \{1, 2, 3, 4, 5\}$ $A \leftrightarrow B$

(ii) $C = \{T, O, P\}$ $D = \{P, O, T\}$ then $C = D$

Ref:

Mk New edition Bk5 page one exercise 1:1

Mk Old pg 1

Lesson two

Sub topic: Joint and disjoint sets

Content: definition of terms

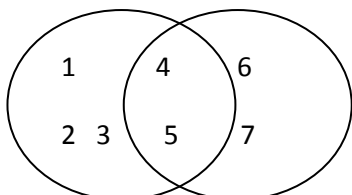
Joint sets have some common elements

Disjoint sets have no common elements

Examples

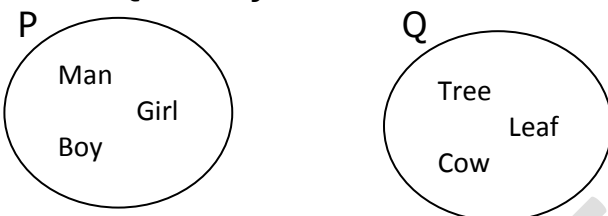
(i) Set $M = \{1, 2, 3, 4, 5\}$ $N = \{4, 5, 6, 7\}$
 $M \cap N = \{4, 5\}$

Set M and N are joint sets



(ii) $P = \{\text{man, boy, girl}\}$ $Q = \{\text{tree, leaf, cow}\}$
 $P \cap Q = \{\}$

P and Q are disjoint sets i.e.



Empty set/Null set

Definitions of terms

Empty set is a set with no member

Symbol for empty set is $\{\}$ or $\emptyset$

Example

$A = \{\text{a car which can fly like a helicopter}\}$ $A = \emptyset$ or $\{\}$

$K = \{\text{animals which lay eggs}\}$ $K = \text{not empty set}$

Union and intersection of sets

Intersection of sets. The symbol used to represent intersection set is $\cap$

Example $A = \{a, b, c, e, f, g\}$, $B = \{b, d, e, f, g\}$

$A \cap B = \{b, e, f, g\}$

Note: common elements must be identified i.e. by circling, ticking or crossing them. This is the main subject competence.

Union of sets the symbol used to represent Union set is $\cup$

Examples $P = \{\bigcirc, \triangle, \square, \bigcirc\}$ $Q = a, b, \triangle, \square$

$P \cup Q = \{\bigcirc, \triangle, \square, a, b\}$

Ref:

Old mk edition bk5 page 3-4

Understanding mtcbk 5 pg 5

Remarks

Lesson three

Sub topic: use of Venn diagrams to represent intersection and union sets

Content: naming parts of a Venn diagram

Elements found in set A

elements found in set B

A only

B only $(B-A)$ or A^1

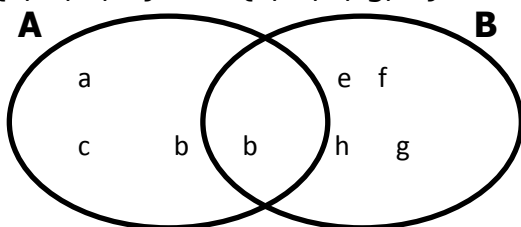
$(A-B)$
Or B^1

Intersection

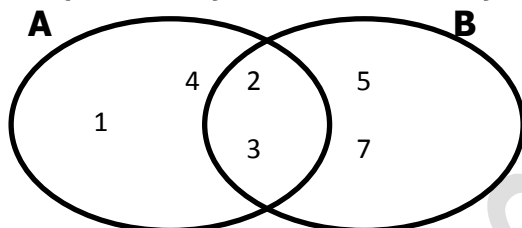
Example: Show the information below on the venn diagram

$A = \{a, b, c, d\}$ $B = \{e, b, f, g, h\}$

$A \cap B = \{b\}$



$A = \{1, 2, 3, 4\}$ $B = \{2, 3, 5, 7\}$



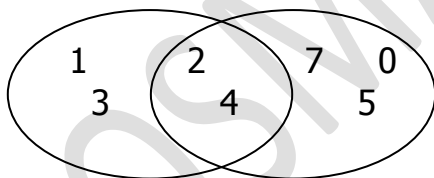
$A \cup B = \{1, 2, 3, 4, 5, 7\}$

$n(A \cup B) = 6$ members

2. Use the venn diagram to answer the questions

X

Y



List the members of set Y

$X = \{2, 4, 7, 0, 5\}$

Find

- i) $X \cap Y$
- ii) $(X \cup Y)$
- iii) $n(X \cup Y) = 7$ elements /members

ref:

Mk new edition bk5 page 5

Mk old edition bk5 page 5

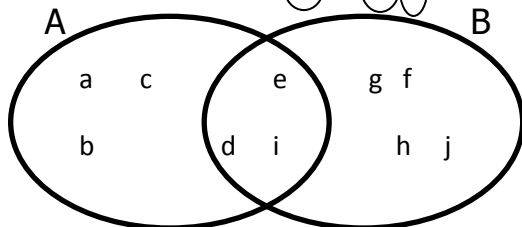
Understanding mtcbk 5 pg 5-6

Remarks

Lesson 4

Sub topic: difference of sets (complements)

Content: $A = \{a, b, \textcircled{i}, c, \textcircled{d}, \textcircled{e}\}$ $B = \{\textcircled{e}, \textcircled{d}, g, f, \textcircled{i}, h, j\}$



- (i) $A - B = \{a, b, c\}$ of $(B)'$
- (ii) $B - A = \{g, f, h, j\}$ or $(A)'$
- (iii) $N(A - B) = 3\text{members}$
- (iv) $N(B - A) = 4\text{members}$

Note: $A - B$ means members in set A only but not in set B (B complement) B^1

$B - A$ means members in set B only but not in set A (A complements) A^1

$B^1 = \{a, b, c\}$

$A^1 = \{g, f, h, j\}$

Ref

Mk new edition 2000 bk5 page 13-14

Mk old bkpg 14-17

Lesson 5

Sub topic: sub sets

Content: definition of terms

A sub set is small set found in a big set

Universal set is a set that contain other smaller sets

Universal set is a subset itself though not a proper subset.

Symbols used

Sub set $\subset$

Not sub set $\not\subset$

Universal set $\in$

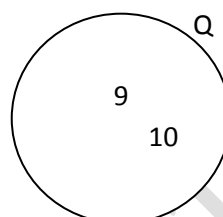
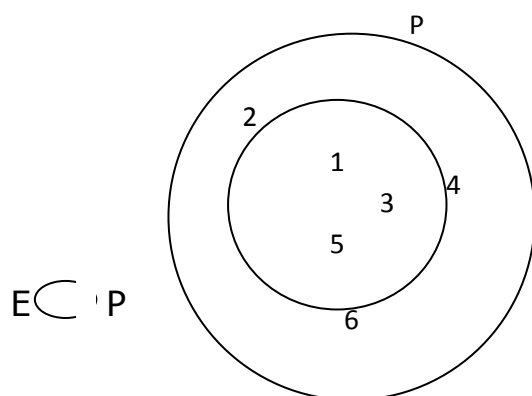
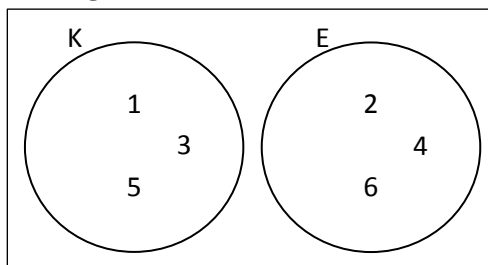
Examples

$P = \{1, 2, 3, 4, 5, 6\}$ $K = \{2, 4, 6\}$ $E = \{1, 3, 5\}$ $Q = \{9, 10\}$

Then

- (i) ECP(E is a sub set of P)
- (ii) $Q \not\subset P$ (Q is not a sub set of P)
- (iii) $P \in$ (P is a universal set of K and E)
- (iv) Represent the given sets on the Venn diagram

$$P = \varepsilon$$



Ref:

Mk old edition exercise 1m book 5 pages 19

Lesson 6

Sub topic: finding the number of sub sets

- (a) By listing
- (b) By use of a formula

Content: examples

Set $K = \{a, b, c\}$

Sub sets of $K = \{a, b, c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a\}, \{b\}, \{c\}, \{\}$

$N(C)K = 8$ sub sets

Using the formula to find the number of elements in set K

$n(\text{C}) = 2^n$ where n stand for number of element is K

2^3

$2 \times 2 \times 2$

8 sub sets

Note:

- (i) Any set is a sub set of itself
- (ii) An empty set is a subset of every set

Shading and describing

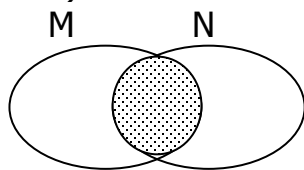
Shaded regions

The curriculum recommends the use of listing method at this level.

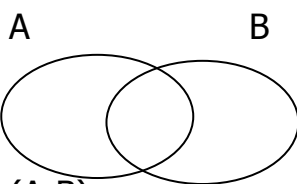
The logical would be that at this level the children have not covered indices and prime factorization.

Examples

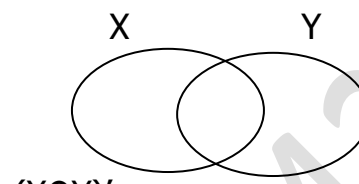
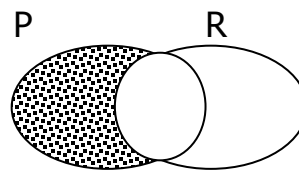
a) Describe the shaded parts



Shade



(A-B)



(X ∩ Y)'

Evaluation activity
New MK bk 5 pg 16

Lesson 8

Sub topic: probability in sets

Content: idea of probability

Probability of zero e.g. sun setting in the north

Probability of 1 e.g. sun setting in the west

Probability of $\frac{1}{2}$ e.g. tossing a coin to get either head or tail

Tossing a coin

Examples: when you toss a coin, what is the probability of a head showing up

Sample space = {head, tail}

$N(S) = 2$

Number of events = (head)
= $n(E) - 1$

$$P = \frac{n(E)}{n(S)} = \frac{1}{2}$$

Toss 2 coins probability of getting two head appearing

Samples = (H.H) (H.T), (T, T), (T, H)

Number of event = $n(E)$ two heads 1

$$P = \frac{n(E)}{n(S)} = \frac{1}{4}$$

Probability of impossibilities occurs when the chances are mutually exclusive.

Ref

Mk old edition bk5 page 22-23

Mk new edition pg 118

Remarks

Lesson 9

Sub topic: tossing a die

Content: examples

What is the chance of 2 appearing when a dice is tossed once?

Sample space = { 1, 2, 3, 4, 5, 6} $n(S) = 6$

No of events = {2} $n(E) = 1$

$$P = \frac{n(E)}{n(S)} = \frac{1}{6}$$

Probability of different items e.g there are 10 pencils in a tin, 3 of them are red and the rest are black, what is the probability of picking a black pencil randomly?

Ref

Mk old edition bk5 exercise 10 page 23

Remarks

Topic two

Topic: Numeration and place value

Sub topic: types of number systems

Lesson one

Content: (a) Hindu and Roman numerals

Hindu	Roman
-------	-------

1	→ I
---	-----

5	→ V
---	-----

10	→ X
----	-----

50	→ L
----	-----

100	→ C
-----	-----

500	→ D
-----	-----

1000	→ M
------	-----

Example	→
---------	---

1. Write 19 in Roman numerals

$$19 = 10 + 9$$

$$= X + IX$$

$$= XIX$$

2. Practice changing 4, 9, 6, 11, 40, 60, 90, 99 etc to Roman numerals and vice versa

Ref:

Mk New edition Bk 5 page 24

Understanding mtcpg 31

Old MK pg 50

Lesson 2

Content: change the given Roman numerals to Hindu Arabic numerals

Example

1. Write XLIX into Hindu Arabic

$$XLIX = XL + IX$$

$$XLIX = 40 + 9$$

$$XLIX = 49$$

2. Practice changing iv, vi, ix, lx, xc, xcix etc to Hindu Arabic numerals and vice versa

Ref

Mk New edition Bk 5 pg 38

Understanding mtchpg 31

MK Old bk 5 pg 50

Remarks:

Lesson 3

Sub topic: addition and subtraction of Roman numerals

Content:

Example

1. Add XXIV + XIX

$$XXV = XX + IV$$

$$XXIV = 20 + 4$$

$$XXIV = 24$$

$$XIX = X + IX$$

$$XIX = 10 + 9$$

$$XIX = 19$$

$$24 + 19 = 43$$

$$43 = XL + III$$

$$43 = XLIII$$

2. Subtract CV – LV

$$CV = C + V$$

$$CV = 100 + 5$$

$$CV = 105$$

$$LV = L + V$$

$$LV = 50 + 5$$

$$LV = 55$$

$$105$$

$$-55$$

$$50 = L$$

Ref

Mk old edition bk 5 pg 53

MK new pg 38

Understanding mtcpg 32

Remarks:

Lesson 4

Sub topic: place value of whole numbers

Content: Writing place value and finding values

Example

H/th	T/th	Th	H	T	O
1	3	4	6	7	8
Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones

The place value of 6 is hundreds

Values of digits in whole numbers

Example

Write the value of each digit in the number 123768

1	2	3	7	6	8	
						Ones = $8 \times 1 = 8$
						Tens = $6 \times 10 = 60$
						Hundreds = $7 \times 100 = 700$
						Thousands = $3 \times 1000 = 3000$
						Ten thousands = $2 \times 10,000 = 20000$
						Hundred thousand = $1 \times 100,000 = 100,000$

- a) Find sum of the place value of 6 and value of 3 in the number 3726
 b) Work out the difference between the place value and value of 8 and 2.

Ref

Old edition pg 30-32

New Mk pg 26-27

Understanding math bk 5 pg 15

Remarks:

Lesson 6

Sub topic: writing figures in words

Content:

Note: we use three zeros '000' to write a thousand

Examples.

Write the following figures in words

- (a) 62 = sixty two
 (b) 108 = one hundred eight
 (c) 9405 = nine thousands four hundred five

Ref

New Mk pg 28

Mk Old Edition Pg 33-34

Understanding mtcbk 5 pg 15

Remarks:

Lesson 6

Sub topic: writing numbers in figures

Content: writing number in figures

Examples

Write in figures

- (a) Four hundred twenty five thousand three hundred seventeen
 Four hundred twenty thousand = 425,000
 Three hundred seventeen = $\begin{array}{r} + \quad 317 \\ \hline 425,317 \end{array}$

Ref

Mk New edition Bk 5 page 29 and Mk old edition page 34
Understandingmtcbkl 5 pg 7

Lesson 7

Sub topic: forming numerals from digits

Content: example

Write down the numbers formed by the digits 3, 7, 5

375, 357, 537, 573, 735, 753

Biggest number formed = 753

Smallest number formed = 357

Note: The biggest number is formed using descending order (big to small)

The smallest number is formed using ascending order (small to big)

Note: use examples with zero as a digit also.

Ref

Mk New edition Bk 5 pg 25 and Mk old edition pg 29

Understanding mtcbk 5 pg 19

Remarks:.....

Lesson 8

Sub topic: expanding whole numbers

(a) Using values

(b) Using place values

(c) Using exponents (powers)

1. Expand 7394 using values = $7000 + 300 + 90 + 4$

2. Expand 3780 using place values: $3780 = (3 \times 1000) + (7 \times 100) + (8 \times 10) + (0 \times 1)$

Sub topic: expanding numbers

(a) Using power of 10 (exponents)

Expand 7914 using powers of ten (10)

$$7914 = (7 \times 10^3) + (9 \times 10^2) + (1 \times 10^1) + (4 \times 10^0)$$

Ref

Mk old edition bk5 pg 39

New mkbk 5 pg 31

Remarks:

Lesson 9

Sub topic: changing from expanded form to single numbers

Content: writing expanded numbers as single numerals

Examples

Write $(4 \times 1000) + (5 \times 100) + (7 \times 10) + (3 \times 1)$

$$\begin{array}{r} 4000 \\ 500 \\ 70 \\ + 3 \\ \hline 4573 \end{array}$$

Ref

Mk new edition bk5 pg 32

Mk Old [g 39-41

Understanding mtcbkpg 33

Lesson 10**SUB TOPIC: ROUNDING OFF WHOLE NUMBERS**

Content:

Examples

1. Round off 53 to the tens

$$\begin{array}{r} 53 \\ +00 \\ \hline 50 \end{array}$$

2. Round off 55 to the tens

$$\begin{array}{r} 55 \\ +10 \\ \hline 60 \end{array}$$

Note: 0, 2, 3, 4, you add 0

5, 6, 7, 8, 9 add the value of the required place value

Ref

Mk new edition bk 5 page 39-44

Mk old 54-55

Understanding mtcpag 20-22

Remarks:

TOPIC THREE

Topic: OPERATION ON WHOLE NUMBERS

Lesson one

Sub topic: Addition of large numbers

Content: addition

Example

$$\begin{array}{r} \text{Add: } 473442 \\ +369215 \\ \hline 842657 \end{array}$$

Masinde went to the market and bought 5 books at 3500/= and 12 pens at 109000/=. How much did he spend altogether?

$$\begin{array}{r} 109000/= \\ + 3500/= \\ \hline 112,500/= \end{array}$$

Ref

Mk New edition Bk5 page 48 -49

MK old edition pg 58-60

Understanding mtcbk 5 pg 36-38

Lesson 2

Sub topic: subtraction of large numbers

Content: subtraction

Example

$$\begin{array}{r} \text{Subtract: } 123643 \\ - 14262 \\ \hline \mathbf{109,381} \end{array}$$

By how much is 367015 greater than 346729?

$$\begin{array}{r} 367015 \\ -346729 \\ \hline \mathbf{20286} \end{array}$$

Ref

Mk New edition Bk5 page 50-57

Understanding mtc pg 40-44

Remarks

Lesson 3

Sub topic: multiplication

Content: multiplication of numbers by one digit

Example

$$450 \times 6$$

$$450$$

$$\begin{array}{r} \times 6 \\ \hline \end{array}$$

$$2700$$

The cost of a book is shs.750/=. Find the cost of 9 similar books at the same rate

	$\begin{array}{r} 750/ = \\ \times \underline{9} \\ \hline 6750/ = \end{array}$ Evaluation activity Mk New edition Bk5 page 52 Mk old pg 53 Understanding mtcbk 5 pg 45-48 Remarks
	Lesson 4 Sub topic: multiplication by two-digit figures Content: example Multiply : $\begin{array}{r} 35 \\ \times 12 \\ \hline 70 \\ +350 \\ \hline 420 \end{array}$ How many pupils are in 33 classrooms if each classroom has 109 pupils? $\begin{array}{r} 109 \\ \times 33 \\ \hline 327 \\ +3270 \\ \hline 3597 \end{array}$ pupils Ref Mk Old edition Bk5 page 64 – 67 Mk new edition bkpg 53-56 Understanding mtcbk 5 pg 46-50 Lesson 5 Sub topic: division of numbers Content: without remainders Example Divide 864 by 6 $\begin{array}{r} 144 \\ 6 \overline{)864} \\ \underline{-6} \\ 26 \\ \underline{-24} \\ 24 \\ \underline{-24} \\ 0 \end{array} = 144$ A school has 480 pupils. Each classroom can take 40 pupils. How many classrooms are there in the school?

Divide 4824 by 12

$$\begin{array}{r} 402 \\ 12 \overline{) 4824} \\ \underline{-48} \\ 002 \\ \underline{-0} \\ 24 \\ \underline{-24} \\ 00 \end{array}$$

25 bottles hold 1725 litres of water, how much does each bottle hold?

Evaluation activity

Mk Old edition Bk5 page 73 and 74 exercise 3N and 3O

Remarks

Lesson 6

Subtopic : Division of numbers

Content: Division with remainder

Examples

i. Divide $12 \div 5$

$$\begin{array}{r} 02 \text{ rem } 2 \\ 5 \overline{) 12} \\ \underline{5 \times 2} \\ 10 \\ \underline{} \\ 2 \end{array}$$

$$12 \div 5 = 2 \frac{2}{5}$$

ii) $126 \div 4$

$$\begin{array}{r} 031 \text{ rem } 2 \\ 4 \overline{) 126} \\ \underline{4 \times 0} \downarrow \\ 12 \\ \underline{4 \times 3} \downarrow \\ 12 \\ \underline{} \downarrow \\ 6 \\ \underline{4 \times 1} \\ 4 \\ \underline{} \\ 2 \end{array}$$

$$126 \div 4 = 31 \frac{2}{4}$$

Ref: New Mk bk 5 pg 58

Old MK bkpg 72-74

Understanding mtcbk 5 pg 57-63

Lesson 7

Sub topic: combined operation of numbers

Content: BODMAS

Example

Workout $\frac{1}{2}$ of $10 + 15 \div 5$

$(\frac{1}{2} \text{ of } 10) + 15 \div 5$

$(\frac{1}{2} \times 10) + 15 \div 5$

$5 + (15 \div 5)$

$5 + 3 = 8$

Ref

Mk New Edition bk 5 page 63

MK old edition pg 75

Lesson 8

Sub topic: statistics

Content: definition of terms

- (a) Mode
- (b) Range
- (c) Median

Example

Given 2, 3, 0, 6, 3 and 4

Find

(a)	Mode	No	Frequency
		0	1
		2	1
		3	2
		4	1
		6	1

Mode = 3

Modal frequency is 2

(b) Range = biggest – smallest

$$6 - 0 = 6$$

(c) Median = 0, 2, 3, 3, 4, 6

$$\begin{array}{r} \frac{3 + 3}{2} = \frac{6}{2} \\ = 3 \end{array}$$

Ref

New Mk pg 64-65

Old MK pg 76

Lesson 9

Sub topic: mean/ average

Content: average = $\frac{\text{Sum of items}}{\text{No of items}}$

Example

Find the average (mean) of 0, 2, and 4

$$\begin{aligned}\text{Average} &= \frac{0+2+4}{3} \\ &= \frac{6}{3} \\ &= 2\end{aligned}$$

Comparing averages and total

The average age of 12 pupils is 9 years. What is their total age?

Average age of 12 is 9

Total age = (12 x 9) years

Total age = 108 years

Ref

Mk Old edition bk5 page 76-79

New MK bk 5 pg 64-65

Remarks

Lesson 10

Sub topic: comparing numbers using symbols

Content: use >, <, =

$$375 \quad \underline{\quad\quad\quad} \quad 752$$

$$5 + 6 \quad \underline{\quad\quad\quad} \quad 6 + 5$$

$$\frac{1}{4} \quad \underline{\quad\quad\quad} \quad \frac{2}{8}$$

Ref

Teacher's collection

New Mk pg 66

Remarks

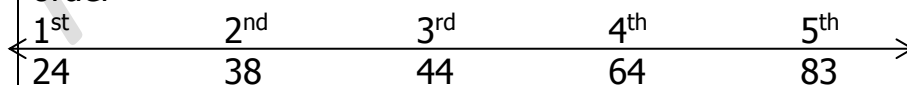
Lesson 11

Sub topic: ordering the numbers on a number line

Content: ascending and descending order

Example

Given 24, 38, 64, 83 and 44 use a number line to arrange the numbers in ascending order



Remarks

Lesson 15

Sub topic: changing to base ten/ decimal base

Content: example

Change 14_{five} to base ten

$$\begin{aligned} 14_{\text{five}} &= (1 \times \text{fives}) + (4 \times \text{ones}) \\ &= (1 \times 5^1) + (4 \times 5^0) = 5 + 4 = 9_{\text{ten}} \end{aligned}$$

Ref

Old MK pg 85

New Mk pg 71

Remarks

Lesson 16

Sub topic: converting base ten to non-decimal bases

Content: example

Change 56_{ten} to base five

	÷	No	Rem
	5	56	1
	5	11	1
		2	

$$= 56_{\text{ten}} = 211_{\text{five}}$$

Ref

Old Mkpg 86

New MK pg 73

Remarks

Lesson 17

Sub topic: addition of numbers in bases (2, 4, and 5)

Content: example

Add $3_{\text{five}} + 4_{\text{five}}$

$$3_{\text{five}} \quad 7 \div 5 = 2 \text{ rem } 1$$

$$\begin{array}{r} + 4_{\text{five}} \\ \hline \end{array}$$

$$\underline{12_{\text{five}}}$$

Ref

Old MK pg 87

New Mk pg 73

Remarks

Lesson 18

Sub topic: subtraction in bases

Content: example

Subtract $123_{\text{five}} - 24_{\text{five}}$

$$\begin{array}{r} 123_{\text{five}} \\ - 24_{\text{five}} \\ \hline \end{array}$$

44_{five}

Ref:

Teacher's collection

Remarks

Lesson 19

Sub topic: multiplication of bases

Content: example

Multiply: $421_{\text{five}} \times 3$

421_{five}

$\times 3_{\text{five}}$

2313_{five}

SDW/side work

$$6 \div 5 = 1 \text{ rem } 1$$

$$13 \div 5 = 2 \text{ rem } 3$$

Note: emphasize should be put on side work.

Ref

Old MK pg 88

New MK pg 74

Remarks:

Lesson 20

Sub topic: finite system

Content: counting in finite five and seven

Example

$$1(\text{finite } 5) = 6, 11, 16, 21, \dots$$

$$3(\text{finite } 5) = 8, 13, 18, 23, \dots$$

Table of finite 5 and 7

Ref:

Old Mk pg 89-91

Lesson 21

Sub topic: addition in finite system (2, 5, 7)

Content: example

$$2 + 3 = \underline{\quad} (\text{finite } 5)$$

$$5 \div 5 = 1 \text{ rem } 0 (\text{finite } 5)$$

$$= 0 (\text{finite } 5)$$

Dial method in addition of finite

Ref:

Old MK pg 92-94

Remarks

Lesson 22

Sub topic: subtraction in finite system (2, 5, 7)

Content: example

$$\text{Subtract } 3 - 4 = \underline{\quad} (\text{finite } 5)$$

$(3 + 5) - 4 = \underline{\hspace{1cm}}$ (finite 5)
 $8 - 4 = 4$ (finite 5)
Dial method $3 - 4 = \underline{\hspace{1cm}}$ (finite 5)
Ref
Teacher's collection

Topic: NUMBER FACTS AND SEQUENCE

Lesson 1

Sub topic: divisibility tests of 2 and 3

Content: any number which ends with an even, digit i.e. 0, 2, 4, 6, 8 is divisible by 2
A number is divisible by 3 if the sum of its digits is divisible by 3

Example

$$144 = 1 + 4 + 4 = 9$$

144 is divisible by 3

Ref

Old Mk pg 68-69

Remarks

Lesson 2

Sub topic: divisibility test of 4, 5 and 10

Content: any number ending with 00 or when the last two digits are divisible by 4 is divisible by 4

Example

320, 100, 1540

Any number ending with 0 or 5 is divisible by 5

Example

220, 540, 725

A number ending with 0 is divisible by 10 e.g. 100, 120, 20

Activity

Teacher's collection

Old MK pg 70

Lesson 3

Sub topic: multiples of numbers

Content: definition of terms

(a) A multiple is a product of two numbers

Example

1. $M_5 = \{5, 10, 15, 20, 25, \dots\}$

2. $M_4 = \{4, 8, 12, 16, \dots\}$

Ref

Old Mk pg 99

New MK pg 79

Remarks

Lesson 4

Sub topic: **Lowest Common Multiples**(LCM/ LCD)

Content: listing method

Ladder method

Example

Find LCM of 4 and 6

$$M_4 = \{4, 8, 12, 16, 20, 24, 28, 32, 36, \dots\}$$

$$M_6 = \{6, 12, 18, 24, 30, 36, \dots\}$$

$$\text{Common multiples} = \{12, 24, 36, \dots\}$$

$$\text{LCM} = 12$$

Note: Common members must be identified.

Ladder method

÷	4	6
2	2	3
2	1	3
3	1	1

$$2 \times 2 \times 3$$

$$4 \times 3 = 12$$

Ref

New Mk pg 80

Old MK pg 100

Remarks

Lesson 5

Sub topic: **Factors of Numbers**

Content: definition

A factor is a number which is multiplied by another number to get a multiple

Example

Multiplication

$$F_{12} \quad 1 \times 12 = 12$$

$$2 \times 6 = 12$$

$$3 \times 4 = 12$$

division

$$12 \div 1 = 12$$

$$12 \div 2 = 6$$

$$12 \div 3 = 4$$

$$F_{12} = \{1, 2, 3, 4, 6, 12\}$$

$$F_{12} = \{1, 2, 3, 4, 6, 12\}$$

Ref

New Mk pg 82

Old Mk pg 102

Lesson six

Sub topic: **Greatest Common Factor (GCF/HCF/HCD)**

Content: GCF and HCF refers to the biggest common factor / divisor

Example: Find the GCF of 12 and 18

F_{12}	F_{18}
$1 \times 12 = 12$	$1 \times 18 = 18$
$2 \times 6 = 12$	$2 \times 9 = 18$
$3 \times 4 = 12$	$3 \times 6 = 18$

Identify the common factors

$$F_{12} = \{1, 2, 3, 4, 6, 12\}$$

$$F_{18} = \{1, 2, 3, 6, 9, 18\}$$

$$CF = \{1, 2, 3, 6\}$$

$$GCF = 6$$

Ref

New Mk pg 82

Old Mk pg 102

Remarks

Lesson seven

Sub topic: **Prime and Composite numbers**

Content: definition

Prime number is a number with only two different factors i.e. 1 and a number itself

Composite number is a number with more than two different factors

Examples

$$13 = 1 \times 13$$

$$F_{13} = \{1, 13\}$$

13 is a prime number

$$4 = 1 \times 4$$

$$4 = 2 \times 2$$

$$F_4 = \{1, 2, 4\}$$

4 is a composite number

Activity

New MK pg 83

Remarks

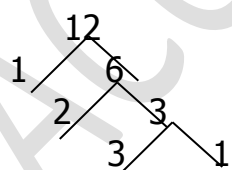
Lesson eight

Sub topic: prime factorization

Content: we use any prime numbers when prime factorizing

Example

Prime factorize



In multiplication form $12 = 2 \times 2 \times 2 \times 3$

In set notation form $12 = 2_1, 2_2, 3_1$.

Note: in set notation form we write small numbers (subscripts) below prime factors when listing them to show the number of times a prime factor has appeared.

In powers form $12 = 2^3 \times 3^1$

Ladder method

$\div$	12
2	6
2	3
3	1

Ref

New MK pg 84-85

Old MK pg 103-105

Remarks

Lesson thirteen

Sub topic: application of square roots

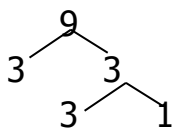
Content: example

If $X^2 = 9$ Find X

$$\sqrt{X^2} = \sqrt{9}$$

$$\sqrt{XXX} = \sqrt{3 \times 3}$$

$$X = 3$$

The area of a square is 16cm^2 . Find the length of one side of the square

$$S \times S = \text{Area}$$

$$S^2 = 16\text{cm}^2$$

$$\sqrt{S \times S} = \sqrt{(2 \times 2) \times (2 \times 2)}$$

$$S = 2 \times 2$$

$$S = 4\text{cm}$$

2	16	
2	8	
2	4	
2	2	
	1	

Lesson 14

Sub topic: set of numbers

Content:

Triangular numbers form triangular patterns when properly arranged

Square numbers are got by multiplying a number by itself

Even numbers are numbers exactly divisible by 2 e.g. 0, 2, 4, 6, 8,

Odd numbers are numbers not exactly divisible by 2 e.g. 1, 3, 5, 7, 9.....

Natural (counting numbers) are numbers used in counting e.g. 1, 2, 3, 4, 5,

Triangular numbers are numbers that form a triangle when arranged

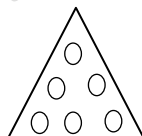
Examples



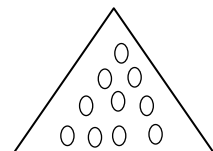
1



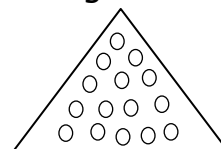
3



6



10



15

Square numbers

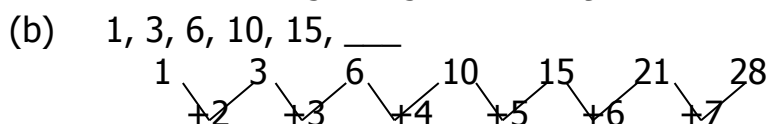
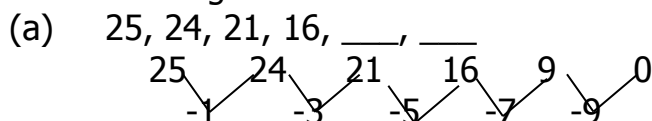
e.g. $\bullet = 1 \times 1$
 $\begin{matrix} \bullet & \bullet \\ \bullet & \bullet \end{matrix} = 4 = 2 \times 2$
 $\begin{matrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{matrix} = 9 = 3 \times 3$

Lesson 15

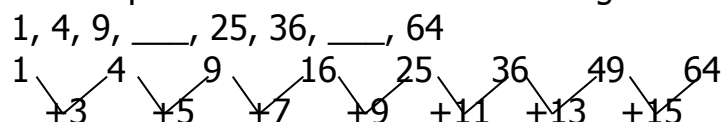
Sub topic: number patterns

Content: example

Fill in the missing numbers



2. Example: find the sum of the missing numbers



Sum = 16 + 49 = 65

Ref

Old MK pg 113-114

New Mk pg 91-92

Remarks

Lesson 16

Sub topic: completing puzzles

Content: magic square

Example

Complete the magic square below

8	a	B
d	5	C
4	e	2

Magic sum = $8 + 5 + 2 = 15$

$d = 15 - (8 + 4) = 15 - 12 = 3$

$c = 15 - (3 + 5) = 15 - 8 = 7$

$b = 15 - (2 + 7) = 15 - 9 = 6$

$e = 15 - (1 + 5) = 15 - 6 = 9$

Ref ; **Understanding mtcpg 89-91**

Remarks

Topic: Fractions

Lesson one

Sub topic: types of fractions

Content:

- (a) Proper fractions (numerator is less than the denominator $\frac{1}{2}$)
- (b) Improper fractions (denominator is less than the numerator $\frac{4}{3}$)
- (c) Mixed fraction (vulgar fractions) includes a whole number and a proper fraction)

- (d) Decimal fractions (numbers with a point)
- (e) Expressing improper fraction as mixed fraction
- (f) Expressing mixed fraction as improper fraction

Example

Express $\frac{9}{5}$ as a mixed number

$$= 5\sqrt{9}$$

$$= \frac{-5}{4}$$

$$= 1 \text{ rem } 4$$

$$= 1\frac{4}{5}$$

Express $1\frac{4}{5}$ as a mixed number

$$= \frac{(WxD) + N}{D}$$

$$= \frac{(1 \times 5) + 4}{5}$$

$$= \frac{5 + 4}{5} = \frac{9}{5}$$

N D

2 12 24

2 6 12

3 3 6

1 2

Ref

Old Mk pg 116-117

New Mk pgpg 115-116

Remarks:

Lesson two

Sub topic: equivalent fractions

Content: examples

$\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$, $\frac{4}{8}$

Ref:

New MK pg 117

Old MK pg 120

Remarks

Lesson three

Sub topic: reducing fractions

Content: example

Reduce $\frac{12}{24}$ to its lowest terms

GCF = 12

$$\frac{12 \div 12}{24 \div 12} = \frac{1}{2}$$

$$\frac{12}{24} = \frac{1}{2}$$

Use only common factors
/divisor

Ref
New Mk pg 118
Old Mk pg 121
Remarks:.....

Lesson four

Sub topic: ordering fractions
Content: using ascending and descending order
Examples : arrange $\frac{1}{3}, \frac{1}{2}, \frac{1}{4}$ in ascending order

LCM = 12

$$\frac{1}{3} \times 12 = 4$$

$$\frac{1}{2} \times 12 = 6$$

$$\frac{1}{4} \times 12 = 3$$

In ascending order $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}$

In descending order $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$

Ref
New MK pg 119
Old Mk pg 125
Remark:.....

Lesson five

Sub topic: comparing fraction using symbols
Content: $>$, $<$, or $=$
Examples which is greater $\frac{1}{3}$ or $\frac{1}{4}$

LCM of 3 and 4 = 12

$$\frac{1}{3} \times 12 = 4$$

$$\frac{1}{4} \times 12 = 3$$

= 4(greater) = 3 (less)

Ref
New MK pg 120
Old Mk pg 126
Remarks:.....

Lesson six

Sub topic: Addition of fractions
Content: different denominations

Examples: Add $\frac{5}{6} + \frac{8}{9} = \frac{(\frac{5}{6} \times 36) + (\frac{8}{9} \times 36)}{36} = \frac{30+24}{36} = \frac{54}{36} = \frac{3}{2} = 1\frac{1}{2}$

Ref

New Mk pg 121
Old MK pg 127
Remarks:.....

Lesson seven

Sub topic: Addition of whole numbers and fractions

Content: Examples: $5 + \frac{3}{4} = \frac{5}{1} + \frac{3}{4}$

$$\text{LCD} = 4$$
$$\frac{\frac{5}{1} \times 4 + \frac{3}{4} \times 4}{4} = \frac{20 + 3}{4} = \frac{23}{4} = 4 \frac{23-20}{3}$$

$$= 5 \frac{3}{4}$$

5 rem 3

Ref:

New Mk pg 122

Old MK pg 128

Remarks

Lesson eight

Sub topic: Addition of mixed numbers

Content: examples

$$\frac{1}{2} + 3 \frac{1}{4}$$

$$3 + (\frac{1}{2} + \frac{1}{4})$$

$$3 + \left(\frac{2+1}{4}\right) = 3 + \frac{3}{4} = 3 \frac{3}{4}$$

Ref

New MK pg 123

Old MK p 129-131

Remarks:.....

Lesson 9

Sub topic: Word problems involving addition of fractions

Content: example

John filled $\frac{1}{2}$ of a tank in the morning and $\frac{2}{5}$ in the afternoon. What fraction of the tank was filled with water?

$$\frac{1}{2} + \frac{2}{5} = \frac{5+4}{10} = \frac{9}{10} \text{ of the tank}$$

Ref

New MK pg125

Old MK pg 131-132

Remarks

Lesson 10

Sub topic: Subtraction of fractions

Content: different denominators

Examples: Subtract $\frac{1}{2} - \frac{1}{3}$ LCM is 6

$$\frac{3-2}{6} = \frac{1}{6}$$

Ref

New MK pg 126-127

Old MK pg 133

Remarks:.....

Lesson 11

Sub topic: Subtraction of fraction from whole numbers

Content: Examples

Subtract $5 - \frac{3}{4}$

$$\frac{5}{1} - \frac{3}{4} = \frac{20-3}{4} = \frac{17}{4} = 4\frac{1}{4}$$

Ref:

New Mk pg 126

Old MK pg 117-118

Remarks:.....

Lesson 12

Sub topic: Subtraction of mixed fractions

Content: Examples

$$4\frac{1}{2} - 1\frac{1}{3} = \frac{9}{2} - \frac{4}{3} = \frac{\frac{9}{2} \times 6 - \frac{4}{3} \times 6}{6} = \frac{27-8}{6} = \frac{19}{6} = 3\frac{1}{6}$$

Ref

New MK pg 126

Old MK pg 133

Lesson 13

Sub topic: Word problems in subtraction of fractions

Content: examples

A baby was given $\frac{5}{6}$ litres of milk and drunk only $\frac{7}{12}$ litres. How much milk remained?

$$\frac{5}{6} - \frac{7}{12} = \frac{\frac{5}{6} \times 12 - \frac{7}{12} \times 12}{12} = \frac{10-7}{12} = \frac{3}{12} \text{ remained}$$

Ref

New Mk pg 127

Old MK pg 134

Remarks:.....

Lesson 14

Sub topic: Combined addition and subtraction

Content: example

Workout:

$$\frac{5}{6} - \frac{5}{9} + \frac{7}{18} = \frac{5}{6} + \frac{7}{18} - \frac{5}{9} = \frac{15 + 7 - 10}{18} = \frac{22 - 10}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$$

Ref

New Mk pg 128

Old Mk pg 135 – 136

Remarks

Lesson 15

Sub topic: Multiplication of whole and fractions

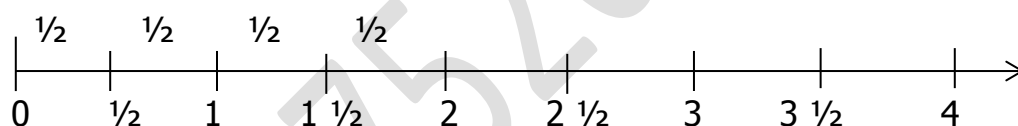
Content: using repeated addition (number line)

Using factor

Example multiply $4 \times \frac{1}{2} = 2$

$$4 \times \frac{1}{2}$$

$$\frac{2 \times 1}{1 \times 2} = 2$$



Ref

New Mk pg 129

Old Mk pg 137

Understanding mtcpg 119

Lesson 16

Sub topic: Multiplication of fractions by a whole

Using “of”

Example simplify: $\frac{1}{2}$ of 16

$$\frac{1}{2} \times 16 = 8$$

Ref

Understanding mtcpg 119-120

New MK pg 129-130

Old Mk pg 137-138

Remarks:.....

Lesson 18

Sub topic: multiplication of unit fraction

Content: example

$$\frac{1}{2} \times \frac{3}{4}$$

$$\frac{1 \times 3}{2 \times 4} = \frac{3}{8}$$

$$2 \times 4 = 8$$

Application of fractions

What is $\frac{1}{4}$ of 1 hour?

$$1 \text{ hr} = 60 \text{ min}$$

$$\frac{1}{4} \text{ hr} = \frac{1}{4} \times 60 \text{ min}$$

$$= 15 \text{ min.}$$

Ref

New MK pg 131

Old MK pg 138

Lesson 18

Sub topic: multiplication of mixed fraction by mixed fraction

Examples

$$1 \frac{1}{2} \times 1 \frac{1}{4} = \frac{3}{2} \times \frac{5}{4} = \frac{3 \times 5}{2 \times 4} = \frac{15}{8} = 1 \frac{7}{8}$$

Ref

Old Mk pg 138

Remarks

Lesson 19

Sub topic: division of fractions

Content: reciprocals of whole numbers

Example

Find the reciprocal of

(a) 2 Let the reciprocal be k

$$2 \times k = 1$$

$$\frac{2k}{2} = \frac{1}{2} = \frac{1}{2}$$

(b) $\frac{1}{4}$ Let the reciprocal be y

$$\frac{1}{4} \times y = 1$$

$$\frac{y}{4} \times 4 = 1 \times 4$$

$$y = 4$$

(c) $1 \frac{1}{3} = \frac{4}{3}$ Let the reciprocal be x

$$\frac{4}{3} \times x = 1$$

$$3 \times \frac{4x}{3} = 1 \times 3$$

$$\frac{4x}{4} = \frac{3}{4}$$

$$x = \frac{3}{4}$$

Note: reciprocal is used instead of upside down

Ref

New Mk pg 131

Old MK pg 141

Remarks:

Lesson 20

Sub topic: Division of wholes by fraction

Content: examples

Workout using reciprocal

$$2 \div \frac{1}{3} = \frac{2}{1} \times \frac{3}{1} = \frac{6}{1} = 6$$

Using the LCM

$$2 \div \frac{1}{3} = \frac{2}{1} \div \frac{1}{3} = \frac{2}{1} \times 3 \div \frac{1}{3} \times 3 = 6 \div 1 = 6$$

Ref

New Mk pg 135

Old Mk pg 142

Remark:.....

Lesson 21

Subtopic: Word problems

Content: examples

(a) How many $\frac{1}{4}$ loaves of bread can be got from 3 loaves of bread?

$$= 3 \text{ loaves} \div \frac{1}{4} \text{ loaves}$$

$$\frac{3}{1} \div \frac{1}{4} = \frac{3}{1} \times \frac{4}{1} = \frac{12}{1} = 12 \text{ quarter loaves}$$

(b) Using LCM and LCM = 4

$$3 \div \frac{1}{4} = \frac{3}{1} \times 4 \div \frac{1}{4} \times 4 = 12 \div 1 = 12 \text{ quarter loaves}$$

Ref

New MK pg 136

Old MK pg 144

Remark.....

Lesson 32

Sub topic: Division of fractions by whole numbers and vice versa

Content: example

(a) Divide $\frac{1}{3} \div 4$

$$\frac{1}{3} \div \frac{4}{1} = \frac{1}{3} \times \frac{1}{4} = \frac{1 \times 1}{3 \times 4} = \frac{1}{12}$$

(b) Divide $4 \div \frac{1}{3}$

$$\frac{4}{1} \div \frac{1}{3} = \frac{4}{1} \times \frac{3}{1} = \frac{4 \times 3}{1 \times 1} = 12$$

Ref

New Mk pg 137-139

Note: give examples on division of mixed fraction and whole number and vice versa

Remarks

Lesson 23

Sub topic: division of a fraction by fraction

Content: example

$$\text{Divide } \frac{2}{3} \div \frac{1}{5} = \frac{2}{3} \times \frac{5}{1} = \frac{10}{3} = 1\frac{1}{3}$$

Old MK pg 144

Teacher's collection

Remarks

Lesson 24

Sub topic: Division of mixed numbers

Content: example

Workout:

$$2\frac{1}{3} \div 3\frac{1}{2} = \frac{7}{3} \div \frac{7}{2} = \frac{7}{3} \times \frac{2}{7} = \frac{14}{21} = \frac{2}{3}$$

Ref

Teacher's collection (see bk6)

TERM II. PRIMARY MATHS NOTES

Topical breakdown

The me	Topic	Sub-topic	Duration	Learning outcome
Numeracy	Fractions	- Converting fractions into decimals and vice versa - Place values of decimals upto hundredths - Finding values of digits in decimals. - Reading and writing decimals in figures and vice versa. - Ordering decimals using a number line / LCM. - Addition and subtraction of decimal numbers - Word problems involving addition and subtraction of decimals.	2 week	The learner is able to solve problems involving decimals related to real life situations.
Geometry	Lines, angles, and geometrical figures	- Construction of; - Parallel lines - Perpendicular lines - Angles - Drawing angles - Measuring angles - Constructing angles (90, 60, 120 only) - Constructing simple shapes using pencils, ruler and a pair of compasses. - Square, rectangle, and equilateral triangle - Lines of folding symmetry - Rectangles - Square - Kites and other shapes - Circles - Construction of hexagons only in a circle	2 weeks	The learner is able to recognize and construct various geometric figures and relate them to other fields such as architectural drawings.
Integration of graphs and data handling	Data handling	- Draw graphs, (bar, picto and line graphs) - Recognize scales on; - On bar graphs - Picto graphs - Line graphs - Interpreting information on graphs - Working out the average of the data.	2weeks	The learner is able to interpret and solve problems related to graphs

		Time	- Telling time on the 12 hour clock only. - Converting hours to minutes and vice versa. - Finding duration in the same time zone. - Finding time, distance and speed. - Solving word problems involving time, distance and speed. - Operation on time (addition and subtraction) - Hours and minutes - Weeks and days	2weeks	The learner is able to apply the knowledge of time in real life situations.
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TOPIC: FRACTIONS

Lesson 1

Sub topic: decimals

Content: place values in figures and words

Examples: what is the place value of each digit in 0.75?

0.75 = 0 . 7 5 6

Ones (1)
 Tenths ($\frac{1}{10}$)
 Hundredths ($\frac{1}{100}$)
 Thousandths ($\frac{1}{1000}$)

REF

Mk New edition Bk5 page 67

Lesson 2

Sub topic: values of digits in decimals

Content: find the value of each digit in 67.253

67.253

Thousandths ($\frac{1}{1000}$) = $3 \times \frac{1}{1000} = 0.003$
 Hundredths ($\frac{1}{100}$) = $5 \times \frac{1}{100} = 0.05$
 Tenths ($\frac{1}{10}$) = $2 \times \frac{1}{10} = \frac{2}{10} = 0.2$
 Ones (1) = $(7 \times 1) = 7$
 Tens (10) = $(6 \times 10) = 60$

Ref

Mk New Edition Bk5 page 68

Old Mk Bk5 page 46

Remarks:.....

Lesson 3

Sub topic: writing decimal fractions in words

Content:

Examples

- (a) Write 0.75 in words

$$0.75 = \frac{75}{100}$$

Seventy five hundredths

- (b) Write 23.137 in words

$$23 \text{ and } \frac{137}{1000}$$

Twenty three and one hundred thirty seven thousandths

Ref.

Old MK pg 46

New MK pg69

Remarks

Lesson 4

Sub topic: writing decimal fraction in figures

Content: Write sixty-three and twenty-five hundredths in figures

$$36 \text{ and } \frac{25}{100}$$

$$63 + 0.25$$

$$63.00$$

$$+0.25$$

$$63.25$$

Activity

New Mk Bk5 page 70

Old mk Bk5 page 47

Remarks

Lesson 5

Sub topic: Expanding decimals

Content: using values

- i) Using values

Examples

Expand 6.25

$$6.25 = 6 + 0.2 + 0.05$$

$$= 6 + \frac{2}{10} + \frac{5}{100}$$

ii) Using powers

$$6.25 = (6 \times 1) + (2 \times 10^{-1}) + (5 \times 10^{-2})$$

Ref

Old MK pg 48-49

New MK pg 36

Remarks:

Lesson 6

Sub topic: Rounding off decimals

Content: round off

0.625 to the nearest tenth

0.625

+ .0

0.6

Round off to the nearest hundredths

10.269

+ 10

10. 27

Ref

Old Mk Maths Bk5 pg 56

Remarks

Lesson 7

Sub topic: decimal fractions

Content: Expressing common fractions as decimals

Example (i) $\frac{1}{1} = 1$ (ii) $\frac{1}{10} = 0.1$ (iii) $\frac{1}{100} = 0.01$

Note: Zero before a decimal point is used to keep the place for the whole number

Ref

Exercise 6:29 and also exercise 5z page 145/ 146 old edition bk5

Remarks

Lesson 8

Sub topic: expressing mixed fractions as decimals

Content: examples

$$3\frac{1}{10} = \frac{(3 \times 10) + 1}{10} = \frac{31}{10} = 3.1$$

Ref

Exercise 6:30 page 142 New Mk Bk5

Exercise 5z page 147 Old Mk Bk5

Remarks

Lesson 9

Sub topic: converting decimals to common fractions

Content: examples

Convert 0.5 to a common fraction

$$0.5 = \frac{5}{10} = \frac{5 \div 2}{10 \div 2} = \frac{1}{2}$$

Ref

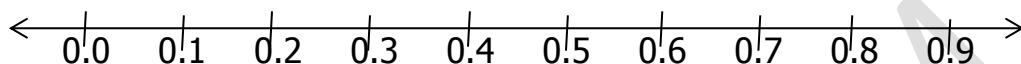
Exercise 6:31 page 143 New Mk Bk5

Lesson 10

Sub topic: comparing decimals using symbols

Content: using symbols $>$, $<$ and $=$

Compare 0.3 _____ 0.5



0.3 > 0.5

Ref

Exercise 3:32 page 145 New Mk Bk5

Exercise from teacher's collection

Remarks

Lesson 11

Sub topic: Ordering decimals

Content: example

Arrange 0.1, 1.1, 0.11 from smallest to greatest and vice versa

$$0.1 = \frac{1}{10}, 0.11 = \frac{11}{100}, 1.1 = \frac{11}{10} \text{ the LCM} = 100$$

$$\frac{1}{10} \times 100 = 10, \frac{11}{100} \times 100 = 11, \frac{11}{10} \times 100 = 110$$

0.1, 0.11, 1.1 ascending order

1.1, 0.11, 0.1 descending order

Ref

Exercise 6:33 page 145-146 New Mk Bk5

Exercise 5z page 149 Old Mk Bk5

Remarks

Lesson 12

Sub topic: addition of decimal fractions

Content: example

Add: 0.45 + 13.2 + 5.2

0.45

13.2

+5.2

18.85

Ref

Exercise 6:34

New Mk pg 77

Remarks

Lesson 13

Sub topic: subtraction of decimal fractions

Content: example

Subtract 13.69 from 97.4

97.4 – 13.69

97.40

-13.69

83.71

Ref

Exercise 6:34

New Mk Bk5pg 79

Remarks

Lesson 14

Sub topic: Addition and subtraction of decimals

Content: example

13.75 – 27 + 91.25

BODMAS

13.75 + 91.25 – 27

13.75

+91.25

105.00

-27.00

78.00

Activity

New MK pg81 / old Mk pg 150

Lesson 15

Sub topic: multiplication of decimals by 10, 100 and 1000

Content: examples

6.25 x 10

$$\frac{625}{100} \times 10 = \frac{625}{10} = 62.5$$

6.25 x 100

$$\frac{625}{100} \times 100 = 625$$

Ref

Exercise 5z page 151 Old Mk Bk5

Remarks

Lesson 16

Sub topic: multiplication of decimals by decimals

Example: multiply $0.06 \times 0.6 = \frac{6}{100} \times \frac{6}{10} = \frac{36}{1000} = 0.036$

Ref

Exercise 5z page 152 Old Mk Bk5

Remarks

Lesson 17

Sub topic: application of decimals in multiplication

Example: One rope measures 4.75metres. How long in metres will 2.5 ropes be if they are joined together?

1 rope measures 4.75m. 2.5ropes measures?

$$4.75m \times 2.5 = \frac{475}{100} \times \frac{25}{10} = \frac{11875m}{1000} = 11.875m$$

$$\begin{array}{r} 475 \\ +25 \\ \hline 2375 \\ +9500 \\ \hline 11875 \end{array}$$

Ref

Exercise 5z page 153 to 154 old edition bk5

Remarks

Lesson 18

Sub topic: Division of decimals

Content: examples

$$\text{Divide: } 0.12 \div 0.6 = \frac{12}{100} \div \frac{6}{10} = \frac{12}{100} \times \frac{10}{6} = \frac{2}{10} = 0.2$$

Division of decimals by whole number and vice versa

$$0.12 \div 6 = \frac{12}{100} \div \frac{6}{1} = \frac{12}{100} \times \frac{1}{6} = 0.02$$

$$6 \div 0.12 = \frac{6}{1} \div \frac{12}{100} = \frac{6}{1} \times \frac{100}{12} = \frac{100}{2} = 50$$

Ref:

Old Mk pg 155

Lesson 19

Sub topic: Application of division of decimals

A tailor uses 1.8m to make a pair of shorts. How many pairs of shorts will he make from 12.6m?

Let the number of pairs be y

$$1.8 \times y = 12.6m$$

$$\frac{18 \times y}{10} = \frac{126m}{10} = \frac{18y}{10} \times 10 = \frac{126}{10} \times 10 = \frac{18y}{18} = \frac{126}{18} = 7 \text{ pairs of shorts}$$

Ref :

Exercise 5z page 156 Old Edition Mk Bk5

GEOMETRY

Lesson one

Sub topic: parallel lines

Content: definition

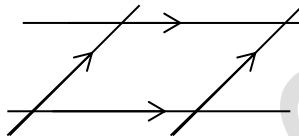
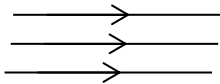
These are lines that are equal distance apart and don't meet when extended in both directions

Drawing parallel lines

Using a ruler



Using ruler and set squares



Ref

Old MK pg 228

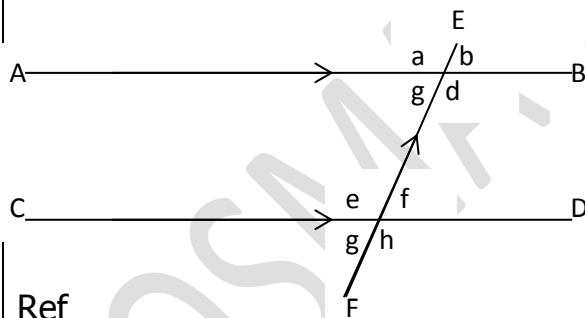
New Mk pg94

Remarks:

Lesson two

Sub topic: intersecting and transversal lines

Content: naming points of intersection lines EF and GH are transversal lines



Ref

New MK pg95

Old Mk pg 231

Remarks:

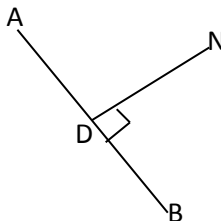
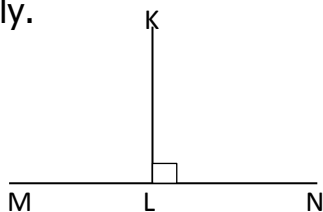
Lesson three

Sub topic: perpendicular lines

Content: definition of perpendicular lines

Naming perpendicular lines from given figures

Drawing or construction of bar lines using pairs of compasses and ruler with pencil only.



KL and ND are perpendicular lines to MN and AB respectively.

Ref

New Mk pg95-96

Lesson four

Sub topic: polygons

Content: naming polygons

Types of triangles

- Equivalent triangles
- Isosceles triangle
- Right angled triangle

Types of quadrilaterals.

- Rectangle
- Square
- Trapezium
- Rhombus
- Kite

Other regular polygons up to 12 sided polygons

Drawing polygons using ruler and pencils (sketches)

Types of polygons

Name	No of sides
Pentagon	5
Hexagon	6
Septagon / heptagon	7
Octagon	8
Nonagon	9
Decagon	10
Nuodecagon	11
Duodecagon	12

Ref:

Old mkbk 5 page 202 exercise 8d

Remarks:

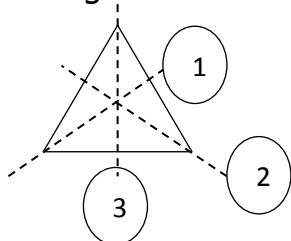
Lesson five

Subtopic: lines of symmetry

Content: defining

Lines of symmetry divide figure into two equal or congruent parts

Drawing and counting the lines of symmetry of i.e. triangles, quadrilaterals e.g.



Nb: child draw and labels

Ref

Old MK pg 231

New mk math bk 5 page 184-185

Remarks:

Lesson six

Sub topic: construction of circles

Content: parts of a circle of different radii and diameter

Drawing circles of radius 3cm

Sub topic: constructing and equilateral triangle in a circle

Content: pupils will use a pair of compasses and a pencil to construct circles equilateral triangles and inscribe

Ref

New Mk pg 186-187

Old Mk pg 250

Lesson seven

Sub topic: Constructing an equilateral triangle without a circle

Example:

Construct an equilateral triangle of side 4cm

Lesson eight

Sub topic: constructing a regular hexagon

Content: pupil will use a pair of compasses and a pencil to construct a regular hexagon in a circle.

Ref

Old Mk pg 251

New MK pg 188

Lesson nine

Sub topic: constructing square in a circle with and without a circle

Content: pupils will construct squares using different radii

Ref

Old MK mtc book 5 pg 252

Lesson ten

Subtopic constructing a rectangle

Content: construction of a rectangle using a pair of compasses

Ref:

Trs' collection

Lesson 11

Sub topic: angles and rotation

Content: definition

Angles is the amount of turning, rotation or opening

Rotation (clockwise or anticlockwise turn through 360°)

Turn clockwise / anticlockwise more through a given angle

Pupils will find the angles that make up turns, half a turn, and a quarter of a turn.

Revolution (a complete turn throughout 360°)

Ref

New MK pg 180-190

Old Mk pg 245-246

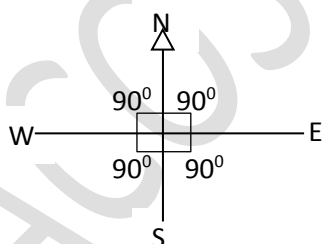
Remarks:

Lesson 12

Sub topic: angles on a compass

Content: pupils will find the different angles between the compass directions

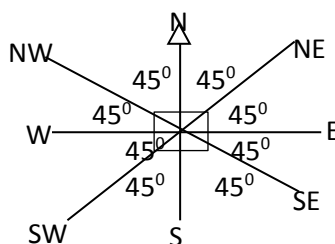
Pupils draw a compass direction



Ref

New MK pg 191

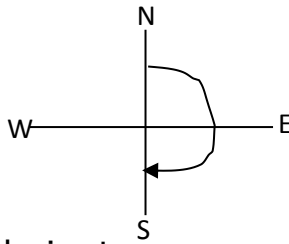
Old MK pg 247



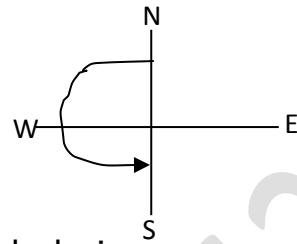
Lesson 13

Sub topic: the clockwise and anticlockwise turns

Content: pupils will find the angles made when one turn clockwise and anticlockwise from the given direction



Clockwise turn...



anticlockwise turn

Examples: Through what angle does Sara turn from North to North East direction in a clockwise direction. Ref

New MK pg 192

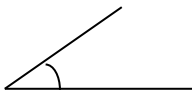
Lesson 14

Sub topic: types of angles

Content: pupils will be guided to name the different types of angles and give examples of such angles

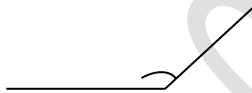
Acute angle, right angle, obtuse angle, straight angle, reflex angle

Acute angle



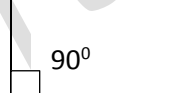
$0^\circ / \angle A / 90^\circ$

obtuse angles



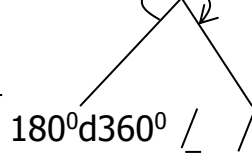
$90^\circ / \angle c / 180^\circ$

right angle



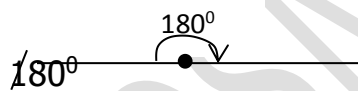
90°

reflex angle



$180^\circ d 360^\circ / /$

Straight angles



Example

Name the types of angles written below

a) 45°

Acute angle

b) 200°

reflex angle

Ref

New Mk bk 5 pg97

Remarks:

Lesson 15

Sub topic: measuring angles using a protractor

Content: pupils will measure different angles using outer scale and inner scale on a protractor with the guidance of the teacher.

Ref

New Mk pg 195

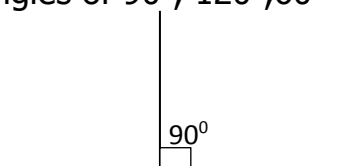
Old MK pg 237

Remarks:

Lesson 16

Sub topic: constructing angles using pair of compasses.

Content: pupils will different angles using paid of compasses, pencil e.g. construct angles of 90° , 120° , 60°



Ref:

New mk math bk 5 pg98

Old MK pg 237

Lesson 17

Sub topic: supplementary angles and complementary angles

Example: what is the supplement of 45°

Let the suple m

$$M + 45^{\circ} = 180^{\circ}$$

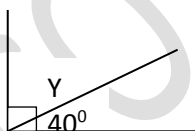
$$M + 45^{\circ} - 45^{\circ} = 180^{\circ} - 45^{\circ}$$

$$M + 0 = 135^{\circ}$$

$$M = 135^{\circ}$$

Complementary angles.

Examples: find the complement of 40°



Let the comp < be Y

$$Y + 40 = 90$$

$$Y + 40 - 40 = 90 - 40$$

$$Y + 0 = 50$$

$$Y = 50^{\circ}$$

Ref

New MK pg102

Old Mk pg 240

Remarks:

Lesson 18

Sub topic: application of complementary and supplementary angles

Content: find complement of 30°

Let the complement be N

$$N + 30^\circ = 90^\circ$$

$$N + 30^\circ - 30^\circ = 90^\circ - 30^\circ$$

$$N + 0 = 60^\circ$$

$$N = 60^\circ$$

The complement of x is 50° . Find the value of x

$$X + 50^\circ = 90^\circ$$

$$X + 50^\circ - 50^\circ = 90^\circ - 50^\circ$$

$$X + 0 = 40^\circ$$

$$X = 40^\circ$$

The supplement of an angle is 72° . What is the angle

let the angle be x

$$X + 72^\circ = 180^\circ$$

$$X + 72^\circ - 72^\circ = 180^\circ - 72^\circ$$

$$X + 0 = 108^\circ$$

$$X = 108^\circ$$

Ref

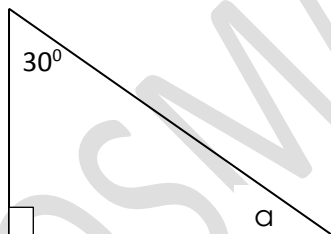
New Mk pg100

Remarks:

Lesson 19

Sub topic: finding angles marked with letters on a triangle

Content: examples find the value of a



$$a + 30^\circ + 90^\circ = 180^\circ$$

$$a + 120^\circ = 180^\circ$$

$$a + 120^\circ - 120^\circ = 180^\circ - 120^\circ$$

$$a + 0 = 60^\circ$$

$$a = 60^\circ$$

Ref

New mk maths bk 5 pg 240

Topic: DATA HANDLING

Lesson one

Sub topic: pictograph interpretation

Content: Pupils will study the given pictograph and workout numbers about the graphs

Musa	  
Mark	 
Jack	   

Key  represents 20 oranges

- (i) How many oranges did Musa get?
1 picture represents 20 oranges
3 pictures represent $20 \times 3 = 60$ oranges
- (ii) How many more oranges did Jack get than Mark?
Jack got $4 \times 20 = 80$ oranges
 $80 \text{ oranges} - 40 \text{ oranges} = 40 \text{ oranges}$
Jack got 40 more oranges than Mark

Ref

New Mk: Maths book 5 pg113-114

Curriculum pg 97-98

Lesson 2:

Sub topic: drawing pictographs

Content: drawing pictographs using the given information and scale

Example

If  represents 10 balls. Draw similar pictures to represent 30 balls

Ref

New MK bk 5 pg 115

Lesson 3:

Sub topic: reading and interpretation of tables

Content: pupils will read and interpret given information then answer questions that follow

Example: Draw the table)

- (i) How many eggs were collected on Tuesday?
10 eggs
- (ii) How many eggs were collected in a week?

- 40 + 10 + 25 + 17 + 53 = 78 + 67 = 145 eggs
- (iii) Find the average number of collected eggs.
- $$\frac{\text{Total}}{\text{No of eggs}} = \frac{145}{5} = 29 \text{ eggs}$$

- (iv) Range
Range = highest – lowest
Range = 53 – 10 = 43 eggs

- (v) Median
- 10 17 25 40 53
-

Ref

New Mk MathsBk 5 pg115

Remarks:

Lesson 5

Sub topic: bar graphs – interpretation

Content: pupils will study given bar graphs and answer the questions that follow

Evaluation activity

New mkmathsbk 5 page 116

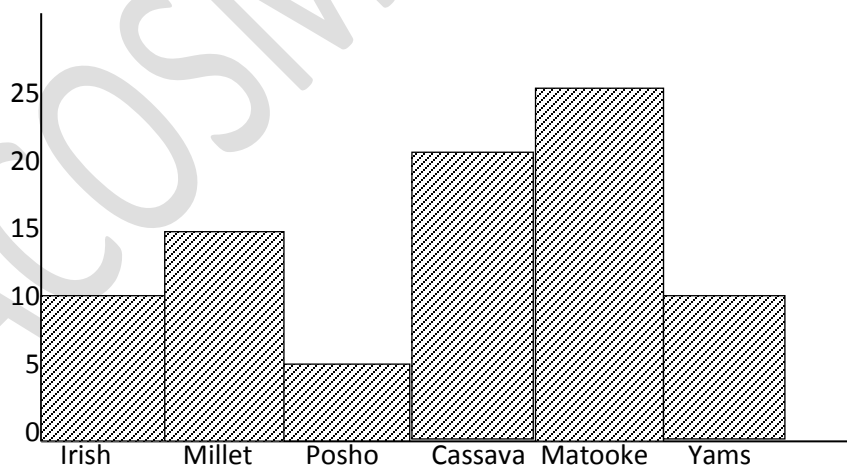
Curriculum pg 97-98

Lesson 6:

Sub topic: drawing bar graphs from tables

Content: pupils will use given tables and scale to draw bar graphs and answer questions that follow

Number of pupils	10	15	5	20	25	10
Types of food	Irish	Millet	Posho	Cassava	Matooke	Yams



Ref

New MK mathsbk 5 pg116-120

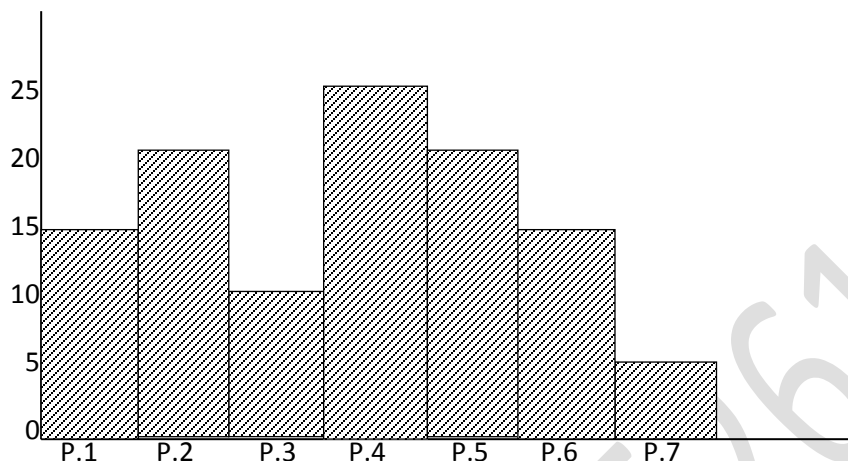
Remarks:

Lesson 7

Sub topic: recording information from a bar graph to a table

Content: pupils will study given bar graphs and record given information on a table

Class	P.1	P.2	P.3	P.4	P.5	P.6	P.7
Number of pupils	15	20	10	25	20	15	5



Ref

New Mk MathsBk 5 Pg 116-120

Teacher guides pupils through example on page 230 and evaluate them

Remarks:

Lesson 8

Sub topic: bar line graphs (interpretation)

Content: pupils will study given bar line graphs and answer the questions that follow

Evaluation activity

New Mk Bk 5 Pg124-127

Remarks:

Lesson 9

Sub topic: drawing bar line graphs

Content: pupils will study given tables and use information to draw bar line graphs

Evaluation activity

Teacher's guidance (do as in bar graph) as in lesson 6 and 7

New Mk Mathsbk 5pg 121-123 exercise 8:16

Remarks

TOPIC: TIME

Lesson 1

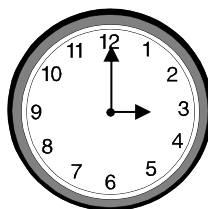
Sub topic: telling time using am and pm (12hour clock system)

Content:

Example

What is the time in

- (a) The morning 3.00am
- (b) The afternoon 3.00pm



Ref

New MK maths bk5 pg129-133

Curriculum pg 98-99

Remarks:.....

Lesson 2

Sub topic: Addition and subtraction of time

Content:

Examples

Add hrs	min	side work
6	25	25
+2	40	40
<u>9</u>	<u>05</u>	<u>65</u>

$\frac{65}{60} = 1.05$

Subtract	hr	min
	34	10
-	22	55
<u></u>	<u>11</u>	<u>15</u>

$60 + 10 = 70$

$\frac{70}{60} = 1.15$

11hours and 15mins

Ref

Tr's collection

Understanding mtc.pg 228-229

Lesson 3

Sub topic: finding duration of time

Content

Mugole started walking from home at 7.15am and reached town at 9:15am. How long did it take him?

Reached	9	15am
Started	-7	15am
He took	<u>2</u>	<u>00</u>

Namata started crying at 7.15am and stopped at 8.00am. How long did it take her?

8 00am 60

-7 15am -15

:45

45 She took 45 minutes

Ref

New mkmaths bk5 pg136

Old mkmaths bk5 pg 219

Remarks:.....

Lesson 4

Sub topic: finding distance

Content:

Example

Find the distance a driver covers in 2hours at a speed of 90km/hr

Distance = speed x time

Distance = 90km/hr x 2hrs

Distance = 180km

Ref

New MK maths bk5 pg138-139

Remarks:.....

Lesson 5

Sub topic: finding time

Content: time = $\frac{\text{distance}}{\text{Speed}}$

Example

Calculate the time taken by a car travelling at 60km/hr to cover a distance of 480km

$$T = \frac{D}{S} = \frac{480km}{\frac{60km}{hr}} = 8hrs$$

Ref

New MK maths bk5 pg140

Remarks:.....

Lesson 6

Sub topic: finding speed

Content

Example

What is the average speed of a cyclist travelling a distance of 150km in 3hours?

$$S = \frac{D}{T} = \frac{150km}{3hrs} = 50km/hr$$

Ref

New MK maths bk5 pg141

Remarks:

MATHEMATICS P.5 LESSON NOTES TERM III

Topical breakdown

Theme	Topic	Sub-topic	Duration	Learning outcome
Measurements	Money	- Recognition of money - Simple rates - Buying and selling (shopping bill) - Table - Listing - Find profits and losses - Cost price and selling price	1 ½	The learner is able to solve practical problems related to utilization of Ugandan currency in everyday life.
	Length, Mass, Capacity	- Conversion of length into cm/ km to m and vice versa. - Calculating perimeter and area of figures i.e. squares, triangles and rectangles only. - Perimeter of a square, triangle and rectangle - Conversion of mass; kg to grams and vice versa. - Solving mathematical problems involving mass. (addition and subtraction) - Conversion of units in capacity. - Solving problems in measurement of capacity. - Addition and subtraction of capacity.	2 ½ weeks	The learner is able to recognize and use standard instruments and units for measuring length, mass and capacity.
Numeracy	Integers	- Drawing numberlines and identifying positive and negative integers - Arranging integers - Comparing integers using symbols $\leq, \geq$ - Addition and subtraction of integers - Mathematical statements and interpreting numberlines. - Solving word problems involving integers.	2 weeks	The learner is able to solve mathematical problems and puzzles using the knowledge of integers.
	Algebra	- Forming algebraic expressions - Collecting like terms - Substitution - Solving equations by (subtracting , adding) - Word problems involving addition and subtraction. - Solving by dividing - Solving by multiplying - Word problems involving division and multiplication - Solving equations involving mixed equations. - Solving equations involving square roots - Application of algebra in (perimeter, area and volume)	2 weeks	The learner is able to solve mathematical problems and puzzles using the knowledge of algebra.

TOPIC: MONEY

Lesson 1

Sub topic: money

Content: denominations

Types of money

Coins, e.g. 50, 100, 200, 500

Notes e.g. 1000, 2000, 5000, 10000, 20000, 50000

Examples

Peter had 3 notes of 1000/= each. How much money did he have?

1 note = 1000/=

3 notes = (3×1000) /=

3 notes = 3000/=

NB: do also calculations on a number of coins and notes of different denominations

REF

Teacher's collections

Lesson 2

Sub topic: buying and selling

Content: using price list

Example

1 book costs 200/= what is the cost of 5 similar books?

1 book = 200/=

5 books = (5×200) /=

5 books = 1000/=

Ref

New MK maths bk 5 pg 143

Old MK pp 222

Lesson 3

Sub topic: buying and selling

Content: more simple rates

Examples

Find the cost of 12 similar books

5 books cost 1000/=

1 bk costs $\frac{1000}{5}$

1 bk = 200/=

12 bks costs (200×12)

12 bks costs 2400/=

Ref

New MK pg 239

Old MK pg 222

Lesson 4

Sub topic: shopping bills and change

Content:

Examples

Kiyaga had 10,000/= he bought 2kg of sugar at shs.1600 per kg, 3bars of sopa at 1000/= each bar, ½ kg of salt at 400/= @ kg

- (a) How much did he spend altogether?
- (b) How much did he spend altogether?
- (c) What was his balance

$$\begin{array}{r} 10,000 \\ - 6400 \\ \hline 3600/= \end{array}$$

Item	Method	Amount
2kg of sugar at 1600/= @	2 x 1600/=	3200/=
3bars of soap at 1000/= @	3 x 1000/=	3000/=
½ kg of salt at 400/= @	½ x 400/=	200/=
Total		6400/=

Ref

New mkmaths bk5 pg145-146

Old MK pg 223

Lesson 5

Sub topic: completing bill tables

Content:

Examples.

A father gave the shopping list below to his children

Item	Quantity	Unit cost	Total
Blue band	½ kg	Shs. 4600 each kg	Shs.2300
Bread	loaves	Shs. 800 each loaf	Shs.2400
Tea leaves	¼ kg	Shs.....@kg	Shs.1500
Sugar	4kg	Shs.1800 @ kg	Shs.....
		Total	Shs.....

Complete the shopping bill

Show all the calculations and fill in later and add

Bread

800/= can buy 1 loaf

1/= buys 1/800 x 2400/=

2400/= buy 3 loaves

tea leaves

¼ kg cost 1500/=

1kg costs 1500 ÷ ¼

1kg cost 1500 x 4

= 6000/=

sugar

1kg cost 1800/=

4kg = 1800/=

$$\begin{array}{r} \times 4 \\ \hline 7200/= \end{array}$$

Ref

New mkmathsbk 5 pg145-146

Old MK pg 224

Remarks:

Lesson 6

Sub topic: transport fare

Content:

Example

A taxi driver charges shs5000 for a trip from Kampala to Jinja per person
How much will 7 people pay for the trip?
1 person pays shs.5000/= 5000×7
7 people pay = 5000×7
= 35000/=

Ref

New MK pg 243
Old Mk pg 225-226

Lesson 7

Content: profit and loss

Examples

Andrew bought a goat at 20,000/= and sold it at shs.25000/=. What profit did he make?

Profit = selling price – cost price

Profit = 25000 – 20000

Profit = 5000/=

Matovu bought a goat at 30,000/= and sold it at shs20000/= how much was his loss?

Loss = buying price – selling price

Loss = 30000 – 20000

Loss = 10000/=

Ref

New mkmaths bk5 pg147-149
Curriculum pg 100

Lesson 8

Sub topic: finding cost price using profit and selling price

Content:

Examples

Nambi sold a radio set at 50000/= she made a profit of 10000/=. What was his cost price?

Selling price = 50000/=

Profit = 10000

Cost price = selling price – profit

Cost price = 50000 – 10000

Cost price = 40000/=

Ref

New MK mathsbk 5 pg152

Lesson 9

Sub topic: finding cost price using loss

Content:

Examples

Oketch sold a goat at 15,000 and made a loss of 3000. How much did he buy the goat?

Selling price = 15000

Loss = 3000

Buying price = selling price + loss

Buying price = 15000 + 3000

Buying price = 18000/=

Ref

New mkmathsbk 5 pg151

Remarks:.....

Lesson 10

Sub topic: finding selling using profit and cost price

Content

Examples

A trader bought a shirt at 7500/= and sold it making a profit of shs.3500. what was his selling price?

Buying price shs.7500

Profit = 3500

Selling price = buying price + profit

Selling = 7500 + 3500

Selling price = 11000/=

Ref

New MK maths bk5 pg150-152

Remarks:

Lesson 11

Sub topic: finding selling price using loss

Content:

Examples

A pupil bought a ball at 15000/= and sold it at a loss of 3000/=. What was the selling price of the ball?

Buying price = 15000/=

Loss = 3000/=

Selling price = buying price – loss

Selling price = 15000 – 3000

Selling price = 12000/=

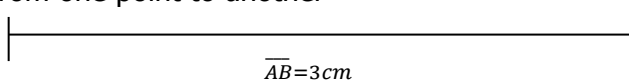
Ref

New MK mathsbk 5 pg150-152

Remarks:.....

Theme : MEASUREMENT**Topic: Length, Mass, Capacity**

Sub topic: length (distance from one point to another)



Content

Estimate in cm and mm

Pupils will measure objects / lines in centimetres and millimetres and record the answers (group activity)

Ref

New MK mathsbk 5 151 and 152

Old MK pg 198

Remarks:

Lesson 2

Subtopic: conversion of metric units

Content: expressing cm to mm and vice versa

Examples

How many mm are 8cm

$$1\text{cm} = 10\text{mm}$$

$$8\text{cm} = (8 \times 10)\text{mm}$$

$$8\text{cm} = 80\text{mm}$$

Convert 120mm to cm

$$10\text{mm} = 1\text{cm}$$

$$1\text{mm} = \left(\frac{1}{10}\right)\text{cm}$$

$$120\text{mm} = \left(\frac{1}{10} \times 120\right)\text{cm}$$

$$120\text{mm} = 12\text{cm}$$

Ref

New MK mathsbkpg 157

Remarks:

Lesson 3

Sub topic: conversion of metres to cm and vice versa

Content

Examples 1

Change 5m to cm

$$5\text{m} = 100\text{cm}$$

$$5\text{m} = (5 \times 100)\text{cm}$$

$$5\text{m} = 500\text{cm}$$

Example 2: Express 1.5m to cm

$$1\text{m} = 100\text{cm}$$

$$1.5\text{m} = \left(\frac{15}{10} \times 100\right)\text{cm}$$

$$1.5\text{m} = 150\text{cm}$$

Example 3:

Change 200cm to m

$$100\text{cm} = 1\text{m}$$

$$1\text{cm} = \left(\frac{1}{100}\right)\text{m}$$

$$100$$

$$200\text{cm} \left(\frac{1}{100} \times 200 \right) \text{m}$$

$$200\text{cm} = 2\text{m}$$

Ref

New MK mathsbk 5 pg 157

Old Mk pp 198

Remarks:

Lesson 4

Sub topic: Addition of m and cm

Content

Examples

Add

$$\begin{array}{r} \text{a)} \quad \text{m} \quad \text{cm} \\ 8 \quad 45 \\ + 1 \quad 55 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b)} \quad \text{M} \quad \text{cm} \\ 2 \quad 73 \\ + 3 \quad 13 \\ \hline \end{array}$$

Ref: _____

Understanding MTC bk 5 pg 144-145

Trs' collection

Lesson 5

Subtopic: Subtraction of m and cm

Content :

Examples: subtract

$$\begin{array}{r} \text{a)} \quad \text{M} \quad \text{cm} \\ 4 \quad 93 \\ - 2 \quad 22 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b)} \quad \text{M} \quad \text{cm} \\ 9 \quad 45 \\ - 3 \quad 65 \\ \hline \end{array}$$

Ref:

Understanding mtcbk 5 pg 142-146

Lesson 6

Sub topic: expressing km to m

Content

Example

Express 2km as metres

$$1\text{km} = 1000\text{m}$$

$$2\text{km} = (2 \times 1000)\text{m}$$

$$2\text{km} = 2000\text{m}$$

Change 15km to m

$$1\text{km} = 1000\text{m}$$

$$15\text{km} = (15 \times 1000)\text{m}$$

$$15\text{km} = 15000\text{m}$$

Convert 0.5km to m

$$1\text{km} = 1000\text{m}$$

$$0.5\text{km} = \left(\frac{5}{10} \times 1000\right)\text{m}$$

$$= 5 \times 100\text{m}$$

$$= 500\text{m}$$

Ref

New mkmathsbk 5 pg158 / Old Mk pp 199

Lesson 7

Sub topic: converting metres to km

Content:

Examples

Change 5000m to km

$$1000\text{m} = 1\text{km}$$

$$1\text{m} = \frac{1}{1000}\text{km}$$

$$5000\text{m} = \frac{5000}{1000}\text{km} = 5\text{km}$$

$$5000\text{m} = 5\text{km}$$

Change 16500m to km

$$1\text{m} = \frac{1}{1000}\text{km}$$

$$16500\text{m} = \left(\frac{16500}{1000}\right)\text{km}$$

$$16500\text{m} = \left(\frac{165}{10}\right)\text{km}$$

$$16500\text{m} = 16.5\text{km}$$

Ref

New mkmathsbk 5 pg 156

Old MK pp 199

Remarks:

Lesson 8

Sub topic: comparing units of measures

Content: using >, < or =

Examples

$$60\text{mm} \quad \underline{\quad} \quad 20\text{cm}$$

$$1\text{cm} = 10\text{mm}$$

$$20\text{cm} = (20 \times 10)\text{mm}$$

$$20\text{cm} = 200\text{mm}$$

$$60\text{mm} < 200\text{mm}$$

$$60\text{mm} < 20\text{cm}$$

Do comparison examples with m and cm and vice versa, km and m and vice versa

Ref

New Mk MathsBk 5 Pg 156

Lesson 9

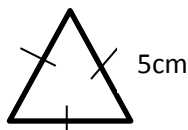
Sub topic: perimeter

Content: finding perimeter of polygons

Regular figures are polygons with all sides equal
Perimeter is the distance around the figure

Example

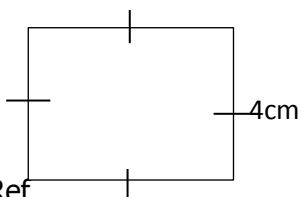
Find the perimeter of the equilateral triangle below



$$\begin{aligned} P &= s + s + s \\ P &= 5 + 5 + 5 \\ P &= 15\text{cm} \end{aligned}$$

Do examples of squares, pentagon, octagons, heptagons etc

Square



$$\begin{aligned} P &= s + s + s + s \\ P &= 4 + 4 + 4 + 4 \\ P &= 8\text{cm} + 8\text{cm} \\ P &= 16\text{cm} \end{aligned}$$

Ref

New Mk mathsbk 5 pg159-161

Old edition Mk pp 203-204

Curriculum pg 101-102

Lesson 10:

Sub topic: finding sides using perimeter

Content:

The perimeter of a square is 12cm. what is the length of each side?

A square has 4sides

$$\frac{4s}{4} = \frac{12}{4} \text{cm}$$

$$s = 3\text{cm}$$

Each side = 3cm

The perimeter of a square is 40cm find the length of each side

A square has four sides

$$P = s + s + s + s$$

$$P = 4s$$

$$\frac{40\text{cm}}{4} = \frac{4s}{4}$$

$$10\text{cm} = s$$

$$S = 10\text{cm}$$

The perimeter of a regular pentagon is 20cm. how long is one of its sides?

A pentagon has 5 sides

$$P = s + s + s + s + s$$

$$\frac{20\text{cm}}{5} = \frac{5s}{5}$$

$$4\text{cm} = s$$

One side = 4cm

Ref

Old MK pp 205-206

New MK pp 284

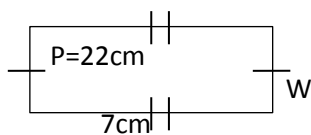
Lesson 11

Sub topic: finding one side of a rectangle using perimeter

Content:

Examples

The perimeter of a rectangle is 22cm and its length is 7cm find its width.

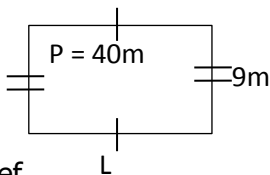


$$P = 2(L + W) \quad 22 - 14 = 14 - 14 + 2W$$

$$22 = 2(7 + W) \quad 8 = 0 + 2W$$

$$22 = 14 + 2W \quad \frac{8}{2} = \frac{2W}{2} = 4cm$$

The perimeter of a rectangle is 40m if its width is 9m find its length



$$P = L + W + L + W \quad 40 - 18 = 2L + 18 - 18$$

$$40 = L + 9 + L + 9 \quad 22 = 2L + 0$$

$$40 = L + L + 9 + 9 \quad \frac{22}{2} = \frac{2L}{2} = 11m$$

$$40 = 2L + 18$$

Ref

New MK pg 284

Old Mk pg 205-206

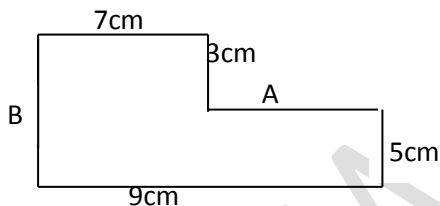
Remarks:

Lesson 12

Sub topic: perimeter of irregular shapes

Content:

Examples



Find the missing sides

Side A

$$A = (9 - 7)cm$$

$$A = 2cm$$

Side B

$$B = 5cm + 3cm$$

$$B = 8cm$$

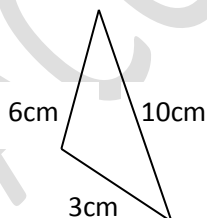
Find the perimeter of the figure

$$P = S + S + S + S + S + S$$

$$P = 7cm + 3cm + 2cm + 5cm + 9cm + 8cm$$

$$P = 34cm$$

Find the perimeter of the scalene triangle below



$$P = S + S + S$$

$$P = 6cm + 3cm + 10cm$$

$$P = 19cm$$

Example 3

Consider

Trapezium

Pentagons

Hexagons

Ref

Teacher's collections and refer to Bk 4

Lesson 13

Sub topic: area of a rectangle

Content

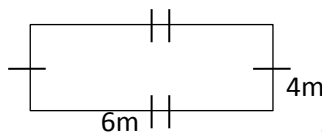
Example

Find the area of the rectangle below

$$A = L \times W$$

$$A = 6m \times 4m$$

$$A = 24m^2.$$



The area of a rectangle is $40dm^2$ and its width is $8dm$. find the length

$$L \times W = 40dm^2$$

$$8 \times L = 40dm^2$$

$$\frac{8 \times L}{8} = \frac{40dm^2}{8} = 5dm^2$$

Ref

Exercise 11:7 pg162-163 Mk new edition / Exercise 8h pg 208 old edition

Lesson 14

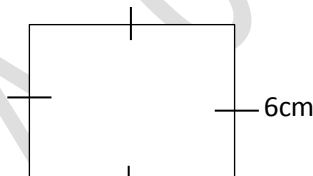
Sub topic: area of a square

Find the area of a square

$$A = S \times S$$

$$A = 6 \times 6$$

$$A = 36cm^2.$$



The area of a square is $36cm^2$ find its sides

$$S \times S = A$$

$$S^2 = A$$

$$\sqrt{S^2} = \sqrt{36cm^2} = 6cm$$

Ref

New Mk mathsBk 5 pg 160 7.9 and pg 281 exercise 12.17

Old MK pg 207

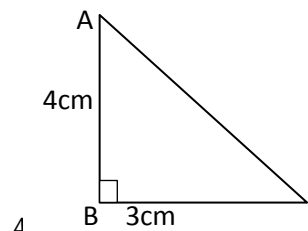
Lesson 15

Sub topic: area of a triangle

Content:

Examples

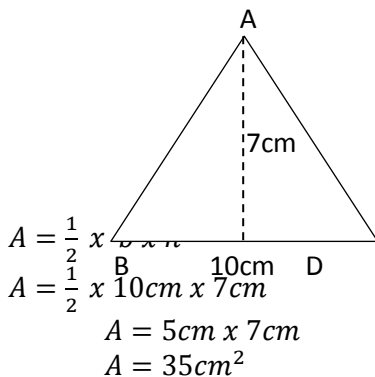
Find the area of the triangles below



$$A = \frac{1}{2} \times 3\text{cm} \times 4\text{cm}$$

$$A = 3\text{cm} \times 2\text{cm}$$

$$A = 6\text{cm}^2$$

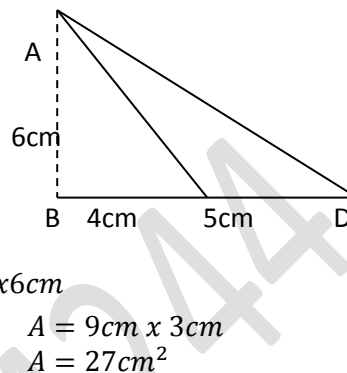


$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times 10\text{cm} \times 7\text{cm}$$

$$A = 5\text{cm} \times 7\text{cm}$$

$$A = 35\text{cm}^2$$



$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times 9\text{cm} \times 6\text{cm}$$

$$A = 9\text{cm} \times 3\text{cm}$$

$$A = 27\text{cm}^2$$

Ref

New MK maths bk5 pg164

Old mk bk5 page 209-210

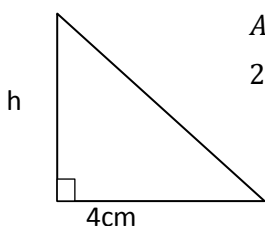
Lesson 16

Sub topic: word problems involving area of triangles

Content:

Examples

The base of a triangle is 4cm and its area is 28cm^2 . Find its height



$$A = \frac{1}{2} \times b \times h$$

$$28\text{cm} = \frac{1}{2} \times 4\text{cm} \times h$$

$$\frac{28}{2} = \frac{2h}{2} = 14$$

Ref

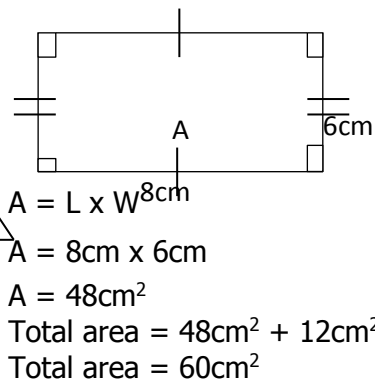
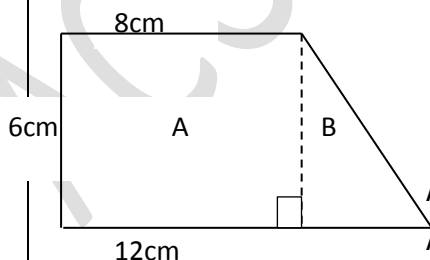
New mk math bk5 pg 163

Lesson 17

Sub topic: area of combined figures

Content:

Find the area of the figures below



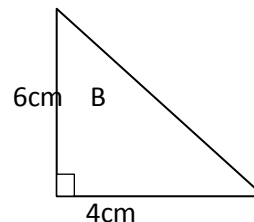
$$A = L \times W$$

$$A = 8\text{cm} \times 6\text{cm}$$

$$A = 48\text{cm}^2$$

$$\text{Total area} = 48\text{cm}^2 + 12\text{cm}^2$$

$$\text{Total area} = 60\text{cm}^2$$



$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times 4 \times 3$$

$$A = 4\text{cm} \times 3\text{cm}$$

$$A = 12\text{cm}^2$$

Ref

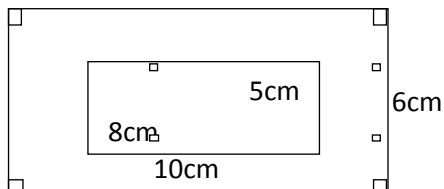
New mkmaths bk5 pg 164-165
Old Mk pp 210-211

Lesson 18

Sub topic: area of shaded and unshaded regions

Content

Examples



$$\begin{aligned} &\text{Area of big rectangle} - \text{area of small rectangle} \\ &= (L \times W) - (L \times W) \\ &= (10 \times 6)\text{cm}^2 - (8 \times 5)\text{cm}^2 \\ &= 60\text{cm}^2 - 40\text{cm}^2 \\ &= 20\text{cm}^2 \end{aligned}$$

Ref

Old mkmaths bk5 pg 212 to 213 exercise 8k

New MK pp 166-167

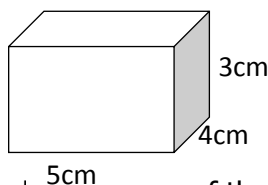
Lesson 19

Sub topic: volume

Content: definition (volume) amount of space inside a container, cubes and cuboids

Examples

Find the volume of the cuboid



Volume

$$V = L \times W \times H$$

$$V = (5 \times 4 \times 3)\text{cm}^3$$

$$V = 60\text{cm}^3$$

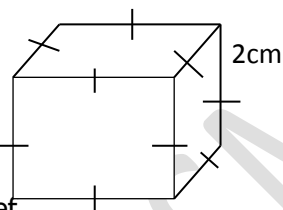
shaded area

$$A = L \times W$$

$$A = (4 \times 3)\text{cm}^2$$

$$A = 12\text{cm}^2$$

Find the volume of the cube below



$$V = S \times S \times S$$

$$V = 2 \times 2 \times 2$$

$$V = 8\text{cm}^3$$

Ref

New MK pp 168-171

Trs' collection

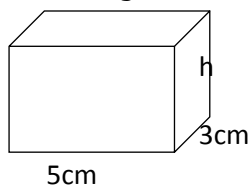
Lesson 20

Sub topic: application of volume

Content:

Examples

Find the missing side of the cuboid given the volume = 50cm^3 .



$$V = L \times W \times h$$

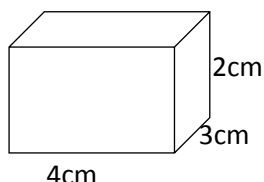
$$60\text{cm}^3 = 5\text{cm} \times 3\text{cm} \times h$$

$$\frac{60\text{cm}^3}{15} = \frac{15\text{cm}^2 h}{15\text{cm}^2}$$

$$4\text{cm} = h$$

Ref

New mk bk5 pg 287 exercise 12.22

Lesson 21**Sub topic: total surface area****Content:****Example****A cuboid has faces**

$$\text{TSA} = 2(L \times W) + 2(L \times h) + 2(h \times W)$$

$$\text{TSA} = 2(4 \times 3) + 2(4 \times 2) + 2(2 \times 3)$$

$$\text{TSA} = 2 \times 12\text{cm}^2 + 2 \times 8\text{cm}^2 + 2 \times 6\text{cm}^2$$

$$\text{TSA} = 24\text{cm}^2 + 16\text{cm}^2 + 12\text{cm}^2$$

$$\text{TSA} = 52\text{cm}^2$$

Ref**Teacher's collection****Lesson 22****Sub topic: capacity****Content: measuring in litres and millilitres**

$$1\text{L} = 1000\text{cm}^3 \text{ or } 1000\text{ml}$$

Examples**Express 5litres of water as****(a) Cubic centimetres**

$$1\text{L} = 1000\text{cm}^3$$

$$5\text{L} = (5 \times 1000)\text{cm}^3$$

$$5\text{L} = 5000\text{cm}^3$$

(b) as millilitres

$$1\text{L} = 1000\text{ML}$$

$$5\text{L} = (5 \times 1000)\text{ML}$$

$$5\text{L} = 5000\text{ML}$$

Ref**New mkbk 5 page 168 exercise 11:12****Lesson 23****Sub topic: comparing metric units****Content: comparing length to weight to capacity****Example**

Place value	Kilo	Hecto	Deca	Basic	Deci	Centi	Milli
Meaning	1000m	100m	10m	Metre gram litre	$\frac{1}{10}$ of m	$\frac{1}{100}$ x m	$\frac{1}{1000}$ x m

Change 3000ML to Litres

$$1000\text{ML} = 1\text{L}$$

$$3000\text{ML} = \frac{3000}{1000}\text{L}$$

$$3000\text{ML} = 3\text{Litres}$$

change 3litres to ML

$$1\text{L} = 1000\text{ML}$$

$$3\text{L} = (3 \times 1000)\text{ML}$$

$$3\text{L} = 3000\text{ML}$$

Ref**New mk math bk5 pg 263 exercise 11.25****New mk math bk 5 page 263 exercise 11:24**

MASS

Lesson 24

Sub topic: expressing grams to kilograms vice versa

Content:

Examples

Change 4000gm to kg

$$1000g = 1kg$$

$$4000g = \left(\frac{4000}{1000}\right)kg$$

$$4000g = 4kg$$

Example 2

Change 3kg to g

$$1kg = 1000g$$

$$3kg = (3 \times 1000)g$$

$$3kg = 3000g$$

Ref

New mkmaths bk5 pg 262 exercise 11.23

Lesson 25

Subtopic: Addition of kg and g

Content

Example 1

$$\begin{array}{r} \text{a)} \quad \text{kg} \quad \text{g} \\ \quad 5 \quad 456 \\ + \quad 2 \quad 204 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b)} \quad \text{kg} \quad \text{g} \\ \quad 4 \quad 596 \\ + \quad 2 \quad 405 \\ \hline \end{array}$$

Ref:

New Mk pp 263

Tr's collection

Subtopic: Subtraction of kg and g

Example

$$\begin{array}{r} \text{a)} \quad \text{kg} \quad \text{g} \\ \quad 8 \quad 765 \\ + \quad 3 \quad 273 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b)} \quad \text{kg} \quad \text{g} \\ \quad 9 \quad 576 \\ + \quad 3 \quad 623 \\ \hline \end{array}$$

Ref:

Tr's collection

Theme: INTEGERS

Lesson 1

Sub topic: Definition

Content:

Integers are numbers represented using a numberline.

(a) Integers – positive and negative numbers including a zero on a numberline.

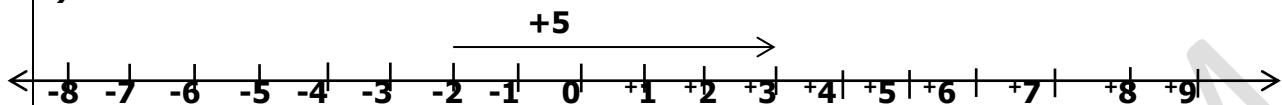
(b) Identifying positive integers

Positive integers have an arrowhead pointing to the right.

Negative integers have an arrowhead pointing to the left.

Examples

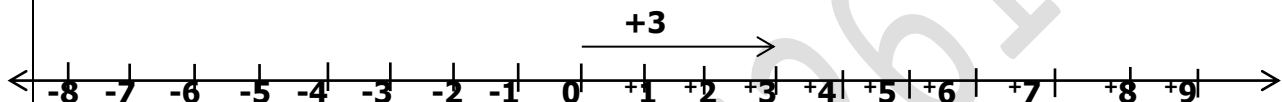
i)



ii)



Example: show +3 on a number line



Ref

Exercise 5: New MK mtc bk5 pg 83-84

Lesson 2

Subtopic: Expressions using integers

Content

- (a) A boy who got no marks in a test is represented by $= 0$.
- (b) A profit of shs 300 - $+300$
- (c) 3 metres below the ground $= -3m$

Ref

Exercise: Class discussion 3 page 96 New MK bk5

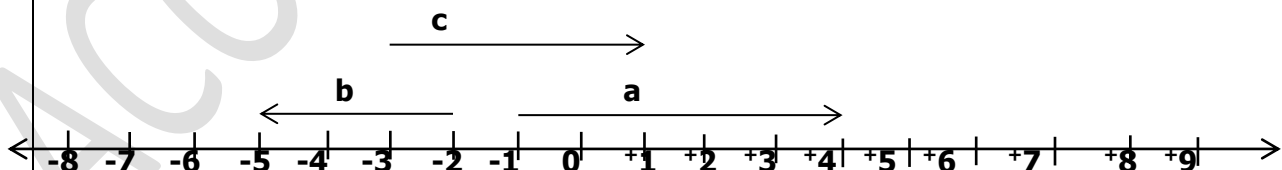
Exercise: Class discussion 2 page 158 old MK bk5

Teachers' collection

Lesson 3

Subtopic: Writing integers represented on a number line

Content:



$a = +5$ $b = -3$ $c = +4$

Ref

New Mk bk5 pg 85

Lesson 4

Subtopic: Comparing integers

Content: comparing integers

Examples: i) Which is smaller -4 or +2?



The one on the left side is always smaller.

$\therefore -4$ is smaller than $+2$

ii) Use $>$, $<$, $=$ to complete

$$+3 > -3$$

Ref

Exercise 6:2 pg86 New MK mtc bk5

Exercise 6e pg 169 old Mtc bk5

Lesson 5

Subtopic: ordering integers

Content: In ascending and descending order

Examples: Arrange -3, +1, -2, 0 and 3 in ascending/ descending orders



$\{-3, -2, 0, +1, +3\}$: ascending order

$\{+3, +1, 0, -2, -3\}$: descending order

Ref:

Exercise 6:4pg85-86

Exercise 6e pg 169 old mtc bk5

Lesson 6

Subtopic: solution sets

Content: Using $>$, $<$, $\geq$, $\leq$

$Y \geq 0$ (means Y are integers greater than or equal to 0)

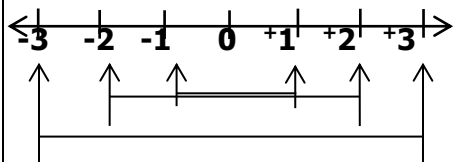


$Y = \{0, +1, +2, +3, +4, +5, +6, \dots\}$

Ref

Exercise 5:3 pg99 New Mtc bk5

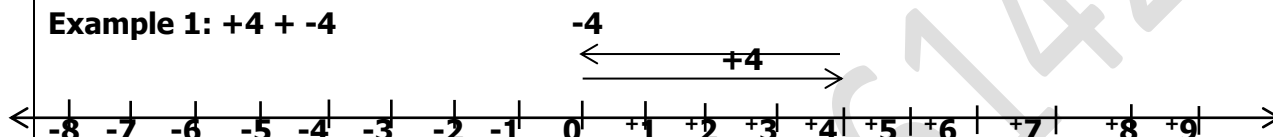
Lesson 7
Subtopic: Inverse of integers
Content: Pairs of inverse



The inverse of -1 is +1
 The inverse of +1 is -1

Additive inverse

Example 1: $+4 + -4$



Note: The additive inverse is a number which gives 0 when added to a number.

Example 2: Calculations

What is the additive inverse of +4:

Let the inverse be x

$$x + 4 = 0$$

$$x + 4 - 4 = 0 - 4$$

$$x + 0 = -4$$

$$x = -4$$

Ref

Exercise 5:4 and 5:5 pages 100 – 102 New MK mtc bk5

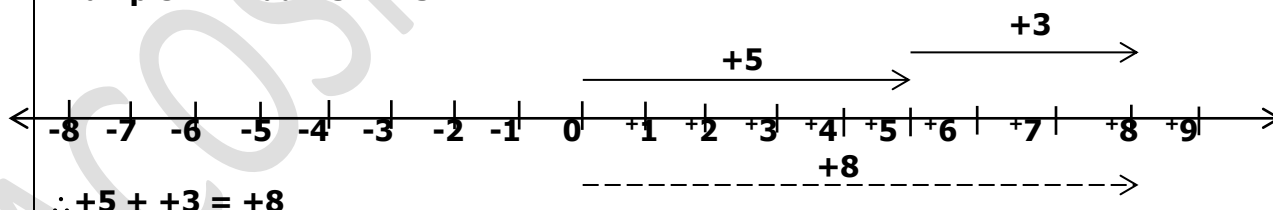
Teachers' collection: Use calculations to find the inverses of 1, -3, 2, +5, 3, -6, 4, x

Lesson 8 (a)

Subtopic: Addition of integers

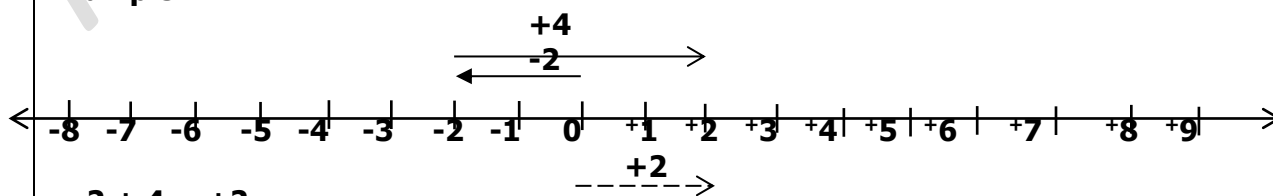
Content: Using a numberline

Example: Add $+5 + +3$



$$\therefore +5 + +3 = +8$$

Example 2



$$\therefore -2 + 4 = +2$$

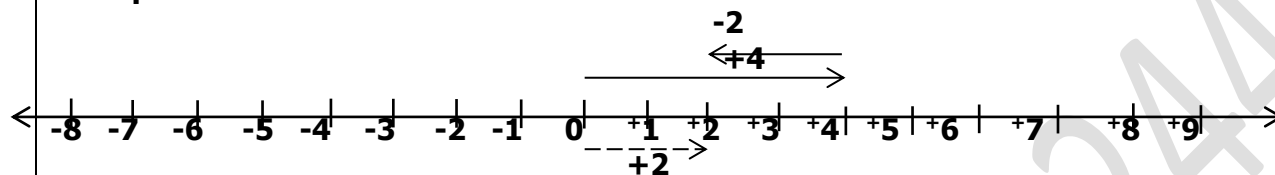
Ref

Exercise 5:6 and 5:7 and 5:8 pg 102 – 104 New MK mtc bk5

Exercise pg 96 OxfordpriMtc bk5 pg96

Lesson 8 (b) Addition of +ve and -ve integers on a number line.

Example: Add $+4 + -2$



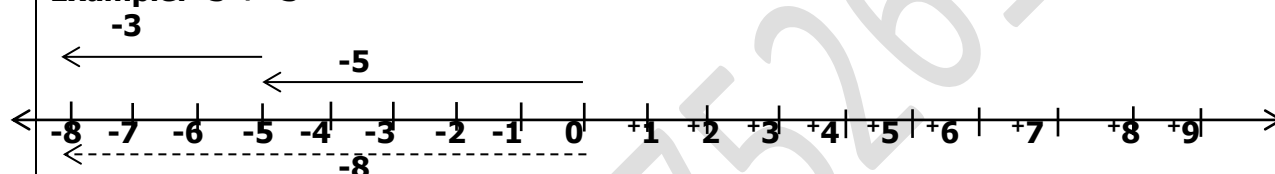
$$\therefore +4 + -2 = +2$$

Exercise 5:7 New Mk edition pg104

NB: Addition of -ve and +ve integers on a numberline

Lesson 8 (c)

Example: $-5 + -3$



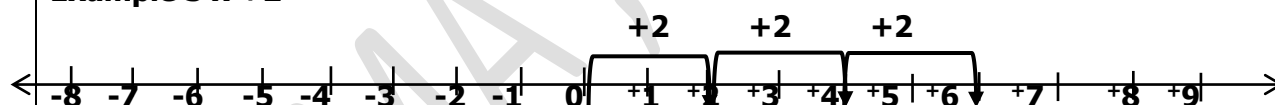
$$\therefore -5 + -3 = -8$$

Ref

New Mk (New edition) pg 104

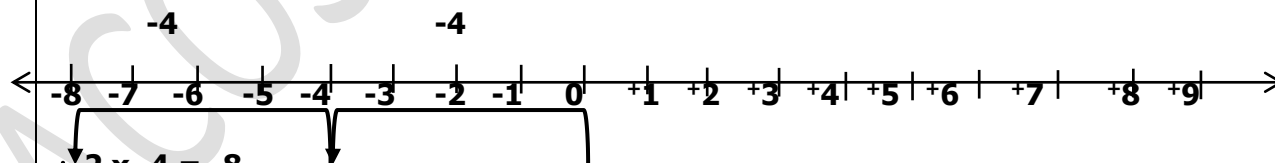
Lesson 8 (d) Multiplication of integers (repeated addition)

Example $3 \times +2$



$$\therefore 3 \times +2 = +6$$

2×-4



$$\therefore 2 \times -4 = -8$$

Ref

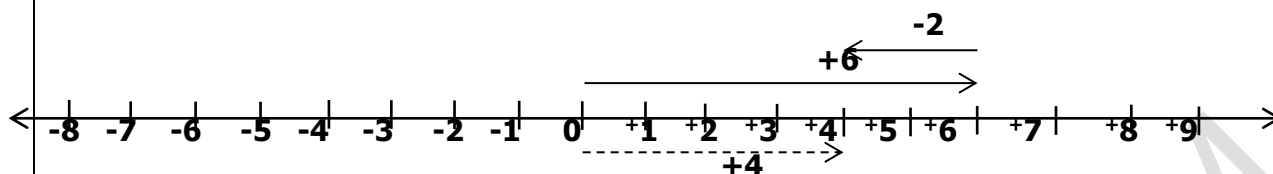
Exercise 8 pg102 Oxford primary Mtc bk5

Trs' collection

Subtraction of integers on a numberline

Lesson 9a: Positive and positive**Example: Subtract $+6 - +2$**

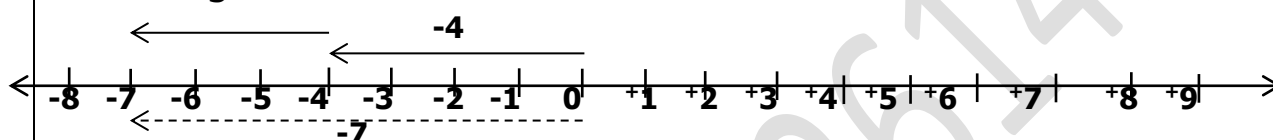
$$= +6 - 2$$



$$+6 - +2 = +4$$

Ref**Exercise 5:15 pg 105-108****Lesson 9b: Negative and positive****Example 1: $-4 - +3 = -4 - +3$**

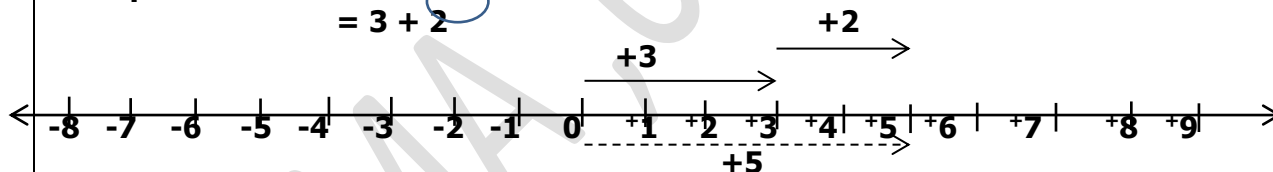
$$= -4 - 3$$



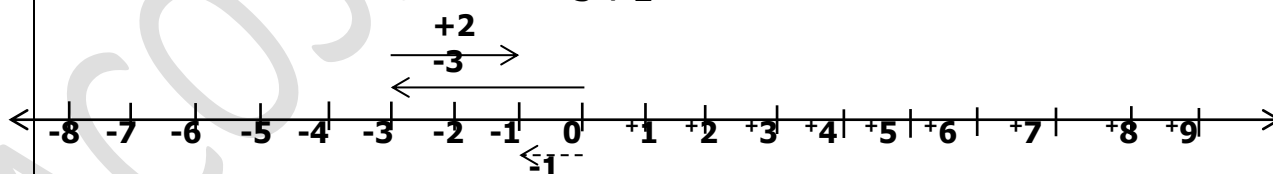
$$-4 - +3 = -7$$

Ref**Exercise 5:9 and 5:10 pgs 105 and 106 new Mtc bk5****Lesson 10a: More subtraction of integers****Content: Positive and negative**

$$\begin{aligned} \text{Example } +3 - -2 &= +3 - -2 \\ &= 3 + 2 \end{aligned}$$

**Lesson 10b: Negative and negative****Example: Subtract $-3 - -2 = -3 - -2$**

$$= -3 + 2$$



$$\therefore -3 - -2 = -1$$

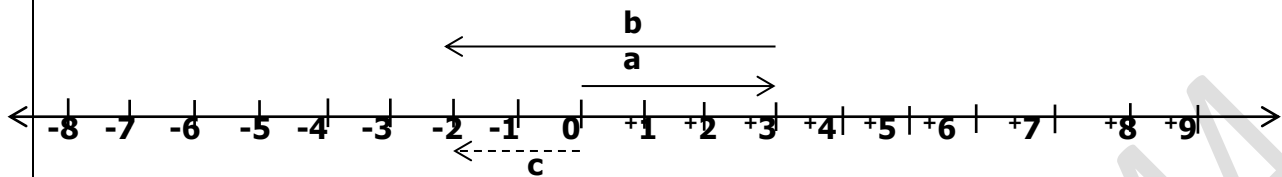
Ref**Exercise 5:11 and 5:12 pg 107 – 108 New Mk bk5**

Lesson 11

Subtopic: Forming mathematical statements

Numberlines

Content: Write the mathematical statement shown on the numberline



$a = +3$, $b = -5$ and $c = -2$

Statement: $+3 + -5 = -2$

Nb: Teach also situation when arrow starts from a -ve side and crosses zero to positive and vice versa

Ref

Exercise 5:13 pg109-110 New MK bk5

Exercise 6c pg106 old edition bk5

Lesson 12a

Subtopic: Addition of integers without using a numberline

Content: Addition

Note:

- i) $(+) + (+) = (+)$
- ii) $(-) + (-) = (-)$
- iii) $(-) + (+) = (-)$ if -ve figure is greater
- iv) $(-) + (+) = (+)$ if +ve figure is greater

Example

Simplify: $+7 + -3$

$$= +7 - 3$$

$$= +4$$

(b) $-3 + -4 = -3 + -4$

$$= -7$$

(d) $-7 + +3$

$$= -7 + 3$$

$$= -4$$

(e) $+3 + +4$

$$+3 + 4$$

$$= +7$$

Ref: 5:15 pg111 New Mk bk5

Lesson 12b

Subtopic: Subtraction of integers without using a numberline

Content note

- i) $(+) - (+) = (-)$ if the 2nd figure is greater
- ii) $(+) - (+) = (+)$ if the 2nd figure is greater
- iii) $(-) - (-) = (+)$ if the 2nd figure is greater
- iv) $(-) - (+) = (-)$

Examples

a) i) $+3 - +7 = 3 - 7 = -4$

ii) $+7 - +3 = 7 - 3 = +4$

b) i) $-3 - -7 = -3 + 7 = +4$

ii) $-7 - -3 = -7 + 3 = -4$

c) i) $-3 - +7 = -3 - 7 = -10$

ii) $-7 - +3 = -7 - 3 = -10$

d) i) $+7 - -3 = +7 + 3 = +10$

ii) $+3 - -7 = +3 + 7 = +10$

ALGEBRA

Lesson 1

Sub topic: forming algebraic expressions

Content

Example

- 4 boys visited my home and later other 2 boys. Later 5 of them left. Form an algebraic equation and simplify it

$$2 \text{ boys} + 4 \text{ boys} - 5 \text{ boys}$$

$$2b + 4b - 5b$$

$$6b - 5b$$

$$= b$$

- A number multiplied by 3 gives 15 let the number be represented by x

$$3x = 15$$

Ref

New MK pp 267-270

Lesson 2

Sub topic: simplifying algebraic expressions

Content

Examples

Write in short

$$q + 7q + 4q = 12q$$

$$4b + 3b - t = 7b - t$$

$$10x - 3x + x$$

$$10x + x - 3x$$

$$11x - 3x = 8x$$

Ref: New MK pp 268

Lesson 3

Sub topic: collecting like terms and simplifying

Content:

Example : collect like terms and simplify

$$4b - 3b + 3t + t$$

$$7y - 8m + y + 10m - 6$$

$$4b - 3b + 3t + t$$

$$7y + y + 10m - 8m - 6$$

$$B + 4t$$

$$8y + 2m - 6$$

Ref

New mkbk 5 pg 269 exercise 12.4

Old Mk pp 174-175

Remarks:

Lesson 4

Sub topic: substitution

Example

If $a = 1$, $b = 3$, $c = 5$

Find the value of $5c + 4b - 8a$

find the value of $\frac{2b}{a+c} = \frac{2 \times 3}{1+5} = \frac{2 \times 3}{1+5} = \frac{6}{6} = 1$

$$(5 \times 5) + (4 \times 3) - (8 - 1)$$

$$25 + 12 - 8$$

$$37 - 8$$

$$29$$

$$abc = a \times b \times c$$

$$abc = 1 \times 3 \times 5$$

$$abc = 3 \times 5$$

$$abc = 15$$

Ref

Exercise 12.6 pg 271 new mk bk5 new edition

MK old edition bk5 pp 177

Lesson 5

Sub topic: solving equations by subtracting

Content

Example

(i) Find the value of a

$$16 + a = 20$$

$$16 - 16 + a = 20 - 16$$

$$0 + a = 4$$

$$a = 4$$

(ii) There are 50 pupils in a class 30 are boys. How many girls are there?

Let the number of girls be g

$$\text{Boys} + \text{girls} = 50$$

$$30 + g = 50$$

$$30 - 30 + g = 50 - 30$$

$$0 + g = 20$$

$$g = 20$$

Ref

New Mk Bk 5 Pg273 exercise 12.8

Old MK pp 179

Lesson 7

Sub topic: solving equations by adding

Content

Example

(1) Solve $n - 5 = 3$

$$n - 5 + 5 = 3 + 5$$

$$n - 0 = 8$$

$$n = 8$$

(2) A boy used 3 of his exercise books and remained with 4 books

How many books did he have at first?

$$B - 3 = 4$$

$$B - 3 + 3 = 4 + 3$$

$B - 0 = 7$
 $B = 7$
 He had 7 books

Ref
 New mk bk5 pg 275 exercise 12.10
 Old MK pp 180
 Remarks:

Lesson 8

Sub topic: solving equations by dividing

Content

Example

(1) Solve $5a = 20$

$$\frac{5a}{5} = \frac{20}{5} = 4$$

(2) The length of a rectangle is 9cm. the width is Ycm. If its area is 72cm^2 find its width.

$L \times W = \text{area}$

$$9\text{cm} \times y = 72\text{cm}^2 \quad \frac{9\text{cm}y}{9\text{cm}} = \frac{72\text{cm}^2}{9\text{cm}}$$

$$Y = 8\text{cm}$$

Ref
 New Mk Bk5 Pg276 exercise 12.11, 12.12
 Old Mk pp 181
 Remarks:

Lesson 10

Sub topic: more equations involving dividing

Content

Solve $x + x + x = 24$

$$3x = 24$$

$$\frac{3x}{3} = \frac{24}{3}$$

$$X = 8$$

solve $2p + 5p = 14$

$$7p = 14$$

$$\frac{7p}{7} = \frac{14}{7}$$

$$p = 2$$

Ref
 New mk bk5 pg 277 exercise 12.13
 Old MK pp 182-183
 Remarks:

Lesson 11

Sub topic: solving equations involving mixed operations

Content

Example

Solve

(a) $4a + 2a + 5 = 23$

$$6a + 5 - 5 = 23 - 5$$

$$6a + 0 = 18$$

$$\frac{6a}{6} = \frac{18}{6}$$

$$a = 3$$

(b) $2x + 5 = 17$

$$2x + 0 = 17 - 5$$

$$2x + 0 = 12$$

$$\frac{2x}{2} = \frac{12}{2}$$

$$x = 6$$

Ref

Newmk bk5 pg 278 exercise 12.14

Lesson 12

Sub topic: equations involving squares

Content: Applying square roots

Example

$$\text{Solve } b^2 = 4$$

$$\sqrt{b^2} = \sqrt{4}$$

$$\sqrt{bxb} = \sqrt{2x2}$$

$$B = 2$$

Ref

New mkbk 5 pg 280 exercise 12.16

Old MK pp 187

Remarks:

Lesson 13:

Sub topic: equations with fractions

Content:

Example

(1) What number when divided by 4 gives 3?

Let the number be x

$$\frac{x}{4} = 3$$

$$\frac{x}{4} = 4x3$$

$$X = 4 \times 3$$

$$X = 12$$

(2) A man divided his money among his three children and each got 450/=. How much money did he give out?

Let the amount of money be represented by m

$$\frac{m}{3} = 450 \neq$$

$$3x \frac{m}{3} = 450x3$$

$$m = 1350 \neq$$

Ref

New MK pp 282-283

Lesson 16

Sub topic: equations involving two fractions

Content:

Example (involving use of LCM)

Find the value of the unknown

$$\frac{3}{5} = \frac{a}{10} \text{ LCM} = 10$$

$$\frac{3}{5} \times 10 = \frac{a}{10} \times 10$$

$$3 \times 2 = a$$

$$a = 6$$

$$\frac{8}{n} = \frac{1}{2} \text{ LCM} = 2n$$

$$\frac{8}{n} \times 2n = \frac{1}{2} \times 2n$$

$$8 \times 2 = n$$

$$n = 16$$

Ref

Exercise 7q pg 185 old mk edition bk5

Remarks:

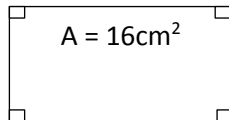
Lesson 17

Sub topic: application of square roots in algebra

Content

Example (Word problems)

The area of a square is 16cm^2 . Find its side



$$s \times s = 16\text{cm}^2$$

$$s^2 = 16\text{cm}^2$$

$$\sqrt{s^2} = \sqrt{16\text{cm}^2}$$

$$s = \sqrt{2 \times 2 \times 2 \times 2} /$$

$$s = 2 \times 2\text{cm}$$

$$s = 4\text{cm}$$

2	16
2	8
2	4
2	2
	1

Ref

Exercise 12.17 pg 281 new edition mkbk 5

Exercise 7x pg 191 old edition mk bk65

Remarks:

Lesson 18

Sub topic: application of algebra (perimeter)

Content

Find the unknown side of a figure when perimeter is given

Example

The perimeter of a square is 36cm find its side in cm

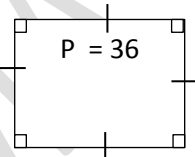
Let side be s

$$s + s + s + s = 36\text{cm}$$

$$4s = 36\text{cm}$$

$$\frac{4s}{4} = \frac{36}{4}$$

$$s = 9\text{cm}$$



||

The perimeter of a rectangle is 40cm. if its length is 15cm. calculate its width

Let the width be represented by w

$$2(L \times W) = P$$

$$2(15\text{cm} + W) = 40\text{cm}$$

$$(2 \times 15\text{cm}) + (2 \times W) = 40\text{cm}$$

$$30\text{cm} + 2W = 40$$

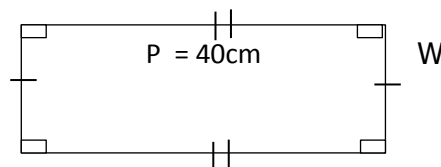
$$30 - 30 + 2W = 40 - 30\text{cm}$$

$$0 + 2W = 10\text{cm}$$

$$\frac{2W}{2} = \frac{10\text{cm}}{2}$$

$$2 = 2$$

$$W = 5$$



Ref

Exercise 12.20 page 284 / 285 New Edition Mk Bk 5

Exercise 7z (ii) page 195 old edition mk bk5

Lesson 19

Sub topic: finding unknown side when given area (rectangle)

Content: rectangle

A long the length

$$7\text{cm}$$

$3x = 15\text{cm}$ (opposite sides of rectangle are equal)

$$3x = 15\text{cm}$$

$$3 \quad 3$$

$$X = 5\text{cm}$$

Along the width

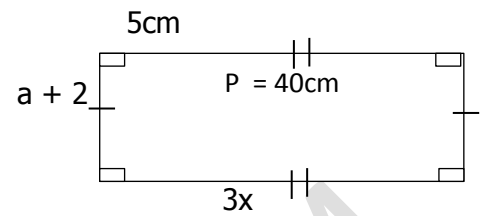
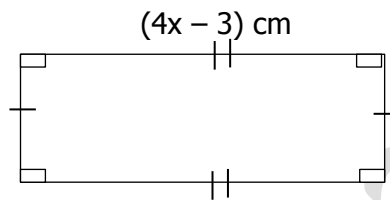
$A + 2 = 7\text{cm}$ (2 opposite sides of a rectangle are equal)

$$A + 2 - 2 = 7 - 5$$

$$A + 0 = 5$$

$$A = 5\text{cm}$$

Find (i) x (ii) length



Ref

Teacher's collections

Lesson 20

Sub topic: finding unknown sides when given area

Content

Example

The area of a rectangle is 32cm^2 its length is 8cm . what is its width?

Let the width be represented by w

$$L \times w = \text{area}$$

$$8\text{cm} \times w = 32\text{cm}^2$$

$$\frac{8\text{cm}W}{8\text{cm}} = \frac{32\text{cm}}{8\text{cm}}$$

$$W = 4\text{cm}$$

Ref

Exercise 12.21 pg 286 new edition mk bk5

Lesson 21

Sub topic: finding unknown sides of cuboids when given volume

Content: example

The volume of a box is 60cm^3 . Its length is 5cm and width is 4cm . find its height

Let h be height

$$L \times W \times h = \text{volume}$$

$$5\text{cm} \times 4\text{cm} \times h = 60\text{cm}^3$$

$$\frac{20\text{cm}^2 h}{20\text{cm}^2} = \frac{60\text{cm}^3}{20\text{cm}^2}$$

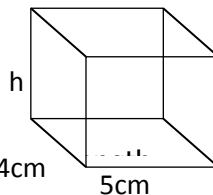
$$H = 3\text{cm}$$

NB: do the same for unknown wid 4cm

Ref

Exercise 12.22 pg 287 new edition MK bk 5

Exercise 7z (iii) pg 196 old edition MK bk 5



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